SOLVED PRACTICE WORKBOOK

General Science: Physics Fundamentals - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

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Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

General Science: Physics Fundamentals - Solved Practice Workbook

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies SI units and dimensional boundary?

A. The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time. B. Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation. C. Work transfers energy when force has a displacement component along its direction; the work-energy theorem links net work to change in kinetic energy, while power is the rate of doing work or transferring energy. Force, energy and power therefore have different dimensions and cannot be interchanged. D. Newton's first law defines inertia, the second links net force to the rate of change of momentum and becomes F = ma for constant mass, and the third pairs equal and opposite forces on different bodies; impulse equals change in momentum and explains why increasing stopping time can reduce average force.

Answer: A. Explanation: The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q2. Which option preserves the technical boundary of SI units and dimensional boundary?

A. Newton's first law defines inertia, the second links net force to the rate of change of momentum and becomes F = ma for constant mass, and the third pairs equal and opposite forces on different bodies; impulse equals change in momentum and explains why increasing stopping time can reduce average force. B. The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time. C. Gravity supplies the centripetal acceleration of an orbiting body; orbital weightlessness is continuous free fall, not absence of gravity. The Basic owner gives Earth's escape velocity as approximately 11.2 km/s and a geostationary orbit as approximately 35,786 km above the equator with one sidereal-day period; escape velocity is independent of projectile mass, while polar or sun-synchronous orbits occupy a different plane and application class. D. Work transfers energy when force has a displacement component along its direction; the work-energy theorem links net work to change in kinetic energy, while power is the rate of doing work or transferring energy. Force, energy and power therefore have different dimensions and cannot be interchanged.

Answer: B. Explanation: The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q3. Which statement uses SI units and dimensional boundary without changing its institution, unit or status?

A. Gravity supplies the centripetal acceleration of an orbiting body; orbital weightlessness is continuous free fall, not absence of gravity. The Basic owner gives Earth's escape velocity as approximately 11.2 km/s and a geostationary orbit as approximately 35,786 km above the equator with one sidereal-day period; escape velocity is independent of projectile mass, while polar or sun-synchronous orbits occupy a different plane and application class. B. Work transfers energy when force has a displacement component along its direction; the work-energy theorem links net work to change in kinetic energy, while power is the rate of doing work or transferring energy. Force, energy and power therefore have different dimensions and cannot be interchanged. C. The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time. D. Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude.

Answer: C. Explanation: The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q4. Which option avoids the standard UPSC close-option trap about SI units and dimensional boundary?

A. Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude. B. Temperature describes thermal state and is not heat; heat is energy transferred because of a temperature difference. Specific heat compares energy required for a temperature rise, while latent heat is absorbed or released during a phase change without a temperature change. C. Gravity supplies the centripetal acceleration of an orbiting body; orbital weightlessness is continuous free fall, not absence of gravity. The Basic owner gives Earth's escape velocity as approximately 11.2 km/s and a geostationary orbit as approximately 35,786 km above the equator with one sidereal-day period; escape velocity is independent of projectile mass, while polar or sun-synchronous orbits occupy a different plane and application class. D. The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time.

Answer: D. Explanation: The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q5. Which statement correctly identifies Motion acceleration and measurement?

A. Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation. B. Work transfers energy when force has a displacement component along its direction; the work-energy theorem links net work to change in kinetic energy, while power is the rate of doing work or transferring energy. Force, energy and power therefore have different dimensions and cannot be interchanged. C. Newton's first law defines inertia, the second links net force to the rate of change of momentum and becomes F = ma for constant mass, and the third pairs equal and opposite forces on different bodies; impulse equals change in momentum and explains why increasing stopping time can reduce average force. D. Gravity supplies the centripetal acceleration of an orbiting body; orbital weightlessness is continuous free fall, not absence of gravity. The Basic owner gives Earth's escape velocity as approximately 11.2 km/s and a geostationary orbit as approximately 35,786 km above the equator with one sidereal-day period; escape velocity is independent of projectile mass, while polar or sun-synchronous orbits occupy a different plane and application class.

Answer: A. Explanation: Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q6. Which option preserves the technical boundary of Motion acceleration and measurement?

A. Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude. B. Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation. C. Gravity supplies the centripetal acceleration of an orbiting body; orbital weightlessness is continuous free fall, not absence of gravity. The Basic owner gives Earth's escape velocity as approximately 11.2 km/s and a geostationary orbit as approximately 35,786 km above the equator with one sidereal-day period; escape velocity is independent of projectile mass, while polar or sun-synchronous orbits occupy a different plane and application class. D. Work transfers energy when force has a displacement component along its direction; the work-energy theorem links net work to change in kinetic energy, while power is the rate of doing work or transferring energy. Force, energy and power therefore have different dimensions and cannot be interchanged.

Answer: B. Explanation: Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q7. Which statement uses Motion acceleration and measurement without changing its institution, unit or status?

A. Temperature describes thermal state and is not heat; heat is energy transferred because of a temperature difference. Specific heat compares energy required for a temperature rise, while latent heat is absorbed or released during a phase change without a temperature change. B. Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude. C. Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation. D. Gravity supplies the centripetal acceleration of an orbiting body; orbital weightlessness is continuous free fall, not absence of gravity. The Basic owner gives Earth's escape velocity as approximately 11.2 km/s and a geostationary orbit as approximately 35,786 km above the equator with one sidereal-day period; escape velocity is independent of projectile mass, while polar or sun-synchronous orbits occupy a different plane and application class.

Answer: C. Explanation: Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q8. Which option avoids the standard UPSC close-option trap about Motion acceleration and measurement?

A. Conduction, convection and radiation are distinct heat-transfer modes, and radiation needs no material medium. The first law applies energy conservation to thermal systems; the second law fixes spontaneous heat flow from hotter to colder bodies and rules out a perfectly efficient heat engine. An ideal black body absorbs and emits across wavelengths, and hotter bodies shift peak emission toward shorter wavelengths. B. Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude. C. Temperature describes thermal state and is not heat; heat is energy transferred because of a temperature difference. Specific heat compares energy required for a temperature rise, while latent heat is absorbed or released during a phase change without a temperature change. D. Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation.

Answer: D. Explanation: Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q9. Which statement correctly identifies Force momentum impulse and Newton laws?

A. Newton's first law defines inertia, the second links net force to the rate of change of momentum and becomes $F = ma$ for constant mass, and the third pairs equal and opposite forces on different bodies; impulse equals change in momentum and explains why increasing stopping time can reduce average force. B. Gravity supplies the centripetal acceleration of an orbiting body; orbital weightlessness is continuous free fall, not absence of gravity. The Basic owner gives Earth's escape velocity as approximately 11.2 km/s and a geostationary orbit as approximately 35,786 km above the equator with one sidereal-day period; escape velocity is independent of projectile mass, while polar or sun-synchronous orbits occupy a different plane and application class. C. Work transfers energy when force has a displacement component along its direction; the work-energy theorem links net work to change in kinetic energy, while power is the rate of doing work or transferring energy. Force, energy and power therefore have different dimensions and cannot be interchanged. D. Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude.

Answer: A. Explanation: Newton's first law defines inertia, the second links net force to the rate of change of momentum and becomes $F = ma$ for constant mass, and the third pairs equal and opposite forces on different bodies; impulse equals change in momentum and explains why increasing stopping time can reduce average force. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q10. Which option preserves the technical boundary of Force momentum impulse and Newton laws?

A. Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude. B. Newton's first law defines inertia, the second links net force to the rate of change of momentum and becomes F = ma for constant mass, and the third pairs equal and opposite forces on different bodies; impulse equals change in momentum and explains why increasing stopping time can reduce average force. C. Gravity supplies the centripetal acceleration of an orbiting body; orbital weightlessness is continuous free fall, not absence of gravity. The Basic owner gives Earth's escape velocity as approximately 11.2 km/s and a geostationary orbit as approximately 35,786 km above the equator with one sidereal-day period; escape velocity is independent of projectile mass, while polar or sun-synchronous orbits occupy a different plane and application class. D. Temperature describes thermal state and is not heat; heat is energy transferred because of a temperature difference. Specific heat compares energy required for a temperature rise, while latent heat is absorbed or released during a phase change without a temperature change.

Answer: B. Explanation: Newton's first law defines inertia, the second links net force to the rate of change of momentum and becomes F = ma for constant mass, and the third pairs equal and opposite forces on different bodies; impulse equals change in momentum and explains why increasing stopping time can reduce average force. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q11. Which statement uses Force momentum impulse and Newton laws without changing its institution, unit or status?

A. Conduction, convection and radiation are distinct heat-transfer modes, and radiation needs no material medium. The first law applies energy conservation to thermal systems; the second law fixes spontaneous heat flow from hotter to colder bodies and rules out a perfectly efficient heat engine. An ideal black body absorbs and emits across wavelengths, and hotter bodies shift peak emission toward shorter wavelengths. B. Temperature describes thermal state and is not heat; heat is energy transferred because of a temperature difference. Specific heat compares energy required for a temperature rise, while latent heat is absorbed or released during a phase change without a temperature change. C. Newton's first law defines inertia, the second links net force to the rate of change of momentum and becomes F = ma for constant mass, and the third pairs equal and opposite forces on different bodies; impulse equals change in momentum and explains why increasing stopping time can reduce average force. D. Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude.

Answer: C. Explanation: Newton's first law defines inertia, the second links net force to the rate of change of momentum and becomes F = ma for constant mass, and the third pairs equal and opposite forces on different bodies; impulse equals change in momentum and explains why increasing stopping time can reduce average force. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q12. Which option avoids the standard UPSC close-option trap about Force momentum impulse and Newton laws?

A. Conduction, convection and radiation are distinct heat-transfer modes, and radiation needs no material medium. The first law applies energy conservation to thermal systems; the second law fixes spontaneous heat flow from hotter to colder bodies and rules out a perfectly efficient heat engine. An ideal black body absorbs and emits across wavelengths, and hotter bodies shift peak emission toward shorter wavelengths. B. A wave transfers energy without net transport of matter, with speed equal to frequency multiplied by wavelength. Sound is mechanical and travels fastest in solids and slowest in gases; frequency primarily sets pitch, amplitude affects loudness, Doppler shift follows relative motion, and resonance amplifies forced oscillation near natural frequency. Infrasound is below 20 Hz, audible sound spans about 20 Hz to 20 kHz and ultrasound is above 20 kHz; ultrasound must not be confused with supersonic motion, while echo and reverberation differ by time separation. C. Temperature describes thermal state and is not heat; heat is energy transferred because of a temperature difference. Specific heat compares energy required for a temperature rise, while latent heat is absorbed or released during a phase change without a temperature change. D. Newton's first law defines inertia, the second links net force to the rate of change of momentum and becomes F = ma for constant mass, and the third pairs equal and opposite forces on different bodies; impulse equals change in momentum and explains why increasing stopping time can reduce average force.

Answer: D. Explanation: Newton's first law defines inertia, the second links net force to the rate of change of momentum and becomes F = ma for constant mass, and the third pairs equal and opposite forces on different bodies; impulse equals change in momentum and explains why increasing stopping time can reduce average force. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q13. Which statement correctly identifies Work energy and power?

A. Work transfers energy when force has a displacement component along its direction; the work-energy theorem links net work to change in kinetic energy, while power is the rate of doing work or transferring energy. Force, energy and power therefore have different dimensions and cannot be interchanged. B. Gravity supplies the centripetal acceleration of an orbiting body; orbital weightlessness is continuous free fall, not absence of gravity. The Basic owner gives Earth's escape velocity as approximately 11.2 km/s and a geostationary orbit as approximately 35,786 km above the equator with one sidereal-day period; escape velocity is independent of projectile mass, while polar or sun-synchronous orbits occupy a different plane and application class. C. Temperature describes thermal state and is not heat; heat is energy transferred because of a temperature difference. Specific heat compares energy required for a temperature rise, while latent heat is absorbed or released during a phase change without a temperature change. D. Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude.

Answer: A. Explanation: Work transfers energy when force has a displacement component along its direction; the work-energy theorem links net work to change in kinetic energy, while power is the rate of doing work or transferring energy. Force, energy and power therefore have different dimensions and cannot be interchanged. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q14. Which option preserves the technical boundary of Work energy and power?

A. Conduction, convection and radiation are distinct heat-transfer modes, and radiation needs no material medium. The first law applies energy conservation to thermal systems; the second law fixes spontaneous heat flow from hotter to colder bodies and rules out a perfectly efficient heat engine. An ideal black body absorbs and emits across wavelengths, and hotter bodies shift peak emission toward shorter wavelengths. B. Work transfers energy when force has a displacement component along its direction; the work-energy theorem links net work to change in kinetic energy, while power is the rate of doing work or transferring energy. Force, energy and power therefore have different dimensions and cannot be interchanged. C. Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude. D. Temperature describes thermal state and is not heat; heat is energy transferred because of a temperature difference. Specific heat compares energy required for a temperature rise, while latent heat is absorbed or released during a phase change without a temperature change.

Answer: B. Explanation: Work transfers energy when force has a displacement component along its direction; the work-energy theorem links net work to change in kinetic energy, while power is the rate of doing work or transferring energy. Force, energy and power therefore have different dimensions and cannot be interchanged. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q15. Which statement uses Work energy and power without changing its institution, unit or status?

A. Conduction, convection and radiation are distinct heat-transfer modes, and radiation needs no material medium. The first law applies energy conservation to thermal systems; the second law fixes spontaneous heat flow from hotter to colder bodies and rules out a perfectly efficient heat engine. An ideal black body absorbs and emits across wavelengths, and hotter bodies shift peak emission toward shorter wavelengths. B. Temperature describes thermal state and is not heat; heat is energy transferred because of a temperature difference. Specific heat compares energy required for a temperature rise, while latent heat is absorbed or released during a phase change without a temperature change. C. Work transfers energy when force has a displacement component along its direction; the work-energy theorem links net work to change in kinetic energy, while power is the rate of doing work or transferring energy. Force, energy and power therefore have different dimensions and cannot be interchanged. D. A wave transfers energy without net transport of matter, with speed equal to frequency multiplied by wavelength. Sound is mechanical and travels fastest in solids and slowest in gases; frequency primarily sets pitch, amplitude affects loudness, Doppler shift follows relative motion, and resonance amplifies forced oscillation near natural frequency. Infrasound is below 20 Hz, audible sound spans about 20 Hz to 20 kHz and ultrasound is above 20 kHz; ultrasound must not be confused with supersonic motion, while echo and reverberation differ by time separation.

Answer: C. Explanation: Work transfers energy when force has a displacement component along its direction; the work-energy theorem links net work to change in kinetic energy, while power is the rate of doing work or transferring energy. Force, energy and power therefore have different dimensions and cannot be interchanged. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q16. Which option avoids the standard UPSC close-option trap about Work energy and power?

A. A wave transfers energy without net transport of matter, with speed equal to frequency multiplied by wavelength. Sound is mechanical and travels fastest in solids and slowest in gases; frequency primarily sets pitch, amplitude affects loudness, Doppler shift follows relative motion, and resonance amplifies forced oscillation near natural frequency. Infrasound is below 20 Hz, audible sound spans about 20 Hz to 20 kHz and ultrasound is above 20 kHz; ultrasound must not be confused with supersonic motion, while echo and reverberation differ by time separation. B. In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence. C. Conduction, convection and radiation are distinct heat-transfer modes, and radiation needs no material medium. The first law applies energy conservation to thermal systems; the second law fixes spontaneous heat flow from hotter to colder bodies and rules out a perfectly efficient heat engine. An ideal black body absorbs and emits across wavelengths, and hotter bodies shift peak emission toward shorter wavelengths. D. Work transfers energy when force has a displacement component along its direction; the work-energy theorem links net work to change in kinetic energy, while power is the rate of doing work or transferring energy. Force, energy and power therefore have different dimensions and cannot be interchanged.

Answer: D. Explanation: Work transfers energy when force has a displacement component along its direction; the work-energy theorem links net work to change in kinetic energy, while power is the rate of

doing work or transferring energy. Force, energy and power therefore have different dimensions and cannot be interchanged. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q17. Which statement correctly identifies Gravitation circular motion and orbits?

A. Gravity supplies the centripetal acceleration of an orbiting body; orbital weightlessness is continuous free fall, not absence of gravity. The Basic owner gives Earth's escape velocity as approximately 11.2 km/s and a geostationary orbit as approximately 35,786 km above the equator with one sidereal-day period; escape velocity is independent of projectile mass, while polar or sun-synchronous orbits occupy a different plane and application class. B. Conduction, convection and radiation are distinct heat-transfer modes, and radiation needs no material medium. The first law applies energy conservation to thermal systems; the second law fixes spontaneous heat flow from hotter to colder bodies and rules out a perfectly efficient heat engine. An ideal black body absorbs and emits across wavelengths, and hotter bodies shift peak emission toward shorter wavelengths. C. Temperature describes thermal state and is not heat; heat is energy transferred because of a temperature difference. Specific heat compares energy required for a temperature rise, while latent heat is absorbed or released during a phase change without a temperature change. D. Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude.

Answer: A. Explanation: Gravity supplies the centripetal acceleration of an orbiting body; orbital weightlessness is continuous free fall, not absence of gravity. The Basic owner gives Earth's escape velocity as approximately 11.2 km/s and a geostationary orbit as approximately 35,786 km above the equator with one sidereal-day period; escape velocity is independent of projectile mass, while polar or sun-synchronous orbits occupy a different plane and application class. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q18. Which option preserves the technical boundary of Gravitation circular motion and orbits?

A. Conduction, convection and radiation are distinct heat-transfer modes, and radiation needs no material medium. The first law applies energy conservation to thermal systems; the second law fixes spontaneous heat flow from hotter to colder bodies and rules out a perfectly efficient heat engine. An ideal black body absorbs and emits across wavelengths, and hotter bodies shift peak emission toward shorter wavelengths. B. Gravity supplies the centripetal acceleration of an orbiting body; orbital weightlessness is continuous free fall, not absence of gravity. The Basic owner gives Earth's escape velocity as approximately 11.2 km/s and a geostationary orbit as approximately 35,786 km above the equator with one sidereal-day period; escape velocity is independent of projectile mass, while polar or sun-synchronous orbits occupy a different plane and application class. C. A wave transfers energy without net transport of matter, with speed equal to frequency multiplied by wavelength. Sound is mechanical and travels fastest in solids and slowest in gases; frequency primarily sets pitch, amplitude affects loudness, Doppler shift follows relative motion, and resonance amplifies forced oscillation near natural frequency. Infrasound is below 20 Hz, audible sound spans about 20 Hz to 20 kHz and ultrasound is above 20 kHz; ultrasound must not be confused with supersonic motion, while echo and reverberation differ by time separation. D. Temperature describes thermal state and is not heat; heat is energy transferred because of a temperature difference. Specific heat compares energy required for a temperature rise, while latent heat is absorbed or released during a phase change without a temperature change.

Answer: B. Explanation: Gravity supplies the centripetal acceleration of an orbiting body; orbital weightlessness is continuous free fall, not absence of gravity. The Basic owner gives Earth's escape velocity as approximately 11.2 km/s and a geostationary orbit as approximately 35,786 km above the equator with one sidereal-day period; escape velocity is independent of projectile mass, while polar or sun-synchronous orbits occupy a different plane and application class. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q19. Which statement uses Gravitation circular motion and orbits without changing its institution, unit or status?

A. Conduction, convection and radiation are distinct heat-transfer modes, and radiation needs no material medium. The first law applies energy conservation to thermal systems; the second law fixes spontaneous heat flow from hotter to colder bodies and rules out a perfectly efficient heat engine. An ideal black body absorbs and emits across wavelengths, and hotter bodies shift peak emission toward shorter wavelengths. B. A wave transfers energy without net transport of matter, with speed equal to frequency multiplied by wavelength. Sound is mechanical and travels fastest in solids and slowest in gases; frequency primarily sets pitch, amplitude affects loudness, Doppler shift follows relative motion, and resonance amplifies forced oscillation near natural frequency. Infrasound is below 20 Hz, audible sound spans about 20 Hz to 20 kHz and ultrasound is above 20 kHz; ultrasound must not be confused with supersonic motion, while echo and reverberation differ by time separation. C. Gravity supplies the centripetal acceleration of an orbiting body; orbital weightlessness is continuous free fall, not absence of gravity. The Basic owner gives Earth's escape velocity as approximately 11.2 km/s and a geostationary orbit as approximately 35,786 km above the equator with one sidereal-day period; escape velocity is independent of projectile mass, while polar or sun-synchronous orbits occupy a different plane and application class. D. In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence.

Answer: C. Explanation: Gravity supplies the centripetal acceleration of an orbiting body; orbital weightlessness is continuous free fall, not absence of gravity. The Basic owner gives Earth's escape velocity as approximately 11.2 km/s and a geostationary orbit as approximately 35,786 km above the equator with one sidereal-day period; escape velocity is independent of projectile mass, while polar or sun-synchronous orbits occupy a different plane and application class. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q20. Which option avoids the standard UPSC close-option trap about Gravitation circular motion and orbits?

A. A wave transfers energy without net transport of matter, with speed equal to frequency multiplied by wavelength. Sound is mechanical and travels fastest in solids and slowest in gases; frequency primarily sets pitch, amplitude affects loudness, Doppler shift follows relative motion, and resonance amplifies forced oscillation near natural frequency. Infrasound is below 20 Hz, audible sound spans about 20 Hz to 20 kHz and ultrasound is above 20 kHz; ultrasound must not be confused with supersonic motion, while echo and reverberation differ by time separation. B. In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence. C. Reflection obeys equality of incidence and reflection angles; plane, concave and convex mirrors form images according to geometry and object position. Mirror action is reflection, so it must not be explained through the refraction mechanism used by lenses. D. Gravity supplies the centripetal acceleration of an orbiting body; orbital weightlessness is continuous free fall, not absence of gravity. The Basic owner gives Earth's escape velocity as approximately 11.2 km/s and a geostationary orbit as approximately 35,786 km above the equator with one sidereal-day period; escape velocity is independent of projectile mass, while polar or sun-synchronous orbits occupy a different plane and application class.

Answer: D. Explanation: Gravity supplies the centripetal acceleration of an orbiting body; orbital weightlessness is continuous free fall, not absence of gravity. The Basic owner gives Earth's escape velocity as approximately 11.2 km/s and a geostationary orbit as approximately 35,786 km above the equator with one sidereal-day period; escape velocity is independent of projectile mass, while polar or sun-synchronous orbits occupy a different plane and application class. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q21. Which statement correctly identifies Pressure buoyancy and fluid principles?

A. Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude. B. Temperature describes thermal state and is not heat; heat is energy transferred because of a temperature difference. Specific heat compares energy required for a temperature rise, while latent heat is absorbed or released during a phase change without a temperature change. C. Conduction, convection and radiation are distinct heat-transfer modes, and radiation needs no material medium. The first law applies energy conservation to thermal systems; the second law fixes spontaneous heat flow from hotter to colder bodies and rules out a perfectly efficient heat engine. An ideal black body absorbs and emits across wavelengths, and hotter bodies shift peak emission toward shorter wavelengths. D. A wave transfers energy without net transport of matter, with speed equal to frequency multiplied by wavelength. Sound is mechanical and travels fastest in solids and slowest in gases; frequency primarily sets pitch, amplitude affects loudness, Doppler shift follows relative motion, and resonance amplifies forced oscillation near natural frequency. Infrasound is below 20 Hz, audible sound spans about 20 Hz to 20 kHz and ultrasound is above 20 kHz; ultrasound must not be confused with supersonic motion, while echo and reverberation differ by time separation.

Answer: A. Explanation: Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q22. Which option preserves the technical boundary of Pressure buoyancy and fluid principles?

A. A wave transfers energy without net transport of matter, with speed equal to frequency multiplied by wavelength. Sound is mechanical and travels fastest in solids and slowest in gases; frequency primarily sets pitch, amplitude affects loudness, Doppler shift follows relative motion, and resonance amplifies forced oscillation near natural frequency. Infrasound is below 20 Hz, audible sound spans about 20 Hz to 20 kHz and ultrasound is above 20 kHz; ultrasound must not be confused with supersonic motion, while echo and reverberation differ by time separation. B. Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude. C. In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence. D. Conduction, convection and radiation are distinct heat-transfer modes, and radiation needs no material medium. The first law applies energy conservation to thermal systems; the second law fixes spontaneous heat flow from hotter to colder bodies and rules out a perfectly efficient heat engine. An ideal black body absorbs and emits across wavelengths, and hotter bodies shift peak emission toward shorter wavelengths.

Answer: B. Explanation: Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q23. Which statement uses Pressure buoyancy and fluid principles without changing its institution, unit or status?

A. A wave transfers energy without net transport of matter, with speed equal to frequency multiplied by wavelength. Sound is mechanical and travels fastest in solids and slowest in gases; frequency primarily sets pitch, amplitude affects loudness, Doppler shift follows relative motion, and resonance amplifies forced oscillation near natural frequency. Infrasound is below 20 Hz, audible sound spans about 20 Hz to 20 kHz and ultrasound is above 20 kHz; ultrasound must not be confused with supersonic motion, while echo and reverberation differ by time separation. B. In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence. C. Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude. D. Reflection obeys equality of incidence and reflection angles; plane, concave and convex mirrors form images according to geometry and object position. Mirror action is reflection, so it must not be explained through the refraction mechanism used by lenses.

Answer: C. Explanation: Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q24. Which option avoids the standard UPSC close-option trap about Pressure buoyancy and fluid principles?

A. In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence. B. Refraction follows a change in light speed across a boundary while frequency remains unchanged; total internal reflection requires travel from denser to rarer medium beyond the critical angle and underpins optical fibres. Convex and concave lenses converge and diverge respectively, with myopia corrected by a concave lens, hypermetropia by a convex lens, presbyopia commonly by bifocals and astigmatism by cylindrical lenses. Dispersion separates colours, Rayleigh scattering favours shorter wavelengths, Mie scattering acts with larger particles and the Tyndall effect reveals colloids. C. Reflection obeys equality of incidence and reflection angles; plane, concave and convex mirrors form images according to geometry and object position. Mirror action is reflection, so it must not be explained through the refraction mechanism used by lenses. D. Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude.

Answer: D. Explanation: Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q25. Which statement correctly identifies Heat temperature specific and latent heat?

A. Temperature describes thermal state and is not heat; heat is energy transferred because of a temperature difference. Specific heat compares energy required for a temperature rise, while latent heat is absorbed or released during a phase change without a temperature change. B. Conduction, convection and radiation are distinct heat-transfer modes, and radiation needs no material medium. The first law applies energy conservation to thermal systems; the second law fixes spontaneous heat flow from hotter to colder bodies and rules out a perfectly efficient heat engine. An ideal black body absorbs and emits across wavelengths, and hotter bodies shift peak emission toward shorter wavelengths. C. A wave transfers energy without net transport of matter, with speed equal to frequency multiplied by wavelength. Sound is mechanical and travels fastest in solids and slowest in gases; frequency primarily sets pitch, amplitude affects loudness, Doppler shift follows relative motion, and resonance amplifies forced oscillation near natural frequency. Infrasound is below 20 Hz, audible sound spans about 20 Hz to 20 kHz and ultrasound is above 20 kHz; ultrasound must not be confused with supersonic motion, while echo and reverberation differ by time separation. D. In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence.

Answer: A. Explanation: Temperature describes thermal state and is not heat; heat is energy transferred because of a temperature difference. Specific heat compares energy required for a temperature rise, while latent heat is absorbed or released during a phase change without a temperature change. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q26. Which option preserves the technical boundary of Heat temperature specific and latent heat?

A. In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence. B. Temperature describes thermal state and is not heat; heat is energy transferred because of a temperature difference. Specific heat compares energy required for a temperature rise, while latent heat is absorbed or released during a phase change without a temperature change. C. Reflection obeys equality of incidence and reflection angles; plane, concave and convex mirrors form images according to geometry and object position. Mirror action is reflection, so it must not be explained through the refraction mechanism used by lenses. D. A wave transfers energy without net transport of matter, with speed equal to frequency multiplied by wavelength. Sound is mechanical and travels fastest in solids and slowest in gases; frequency primarily sets pitch, amplitude affects loudness, Doppler shift follows relative motion, and resonance amplifies forced oscillation near natural frequency. Infrasound is below 20 Hz, audible sound spans about 20 Hz to 20 kHz and ultrasound is above 20 kHz; ultrasound must not be confused with supersonic motion, while echo and reverberation differ by time separation.

Answer: B. Explanation: Temperature describes thermal state and is not heat; heat is energy transferred because of a temperature difference. Specific heat compares energy required for a temperature rise, while latent heat is absorbed or released during a phase change without a temperature change. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q27. Which statement uses Heat temperature specific and latent heat without changing its institution, unit or status?

A. In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence. B. Refraction follows a change in light speed across a boundary while frequency remains unchanged; total internal reflection requires travel from denser to rarer medium beyond the critical angle and underpins optical fibres. Convex and concave lenses converge and diverge respectively, with myopia corrected by a concave lens, hypermetropia by a convex lens, presbyopia commonly by bifocals and astigmatism by cylindrical lenses. Dispersion separates colours, Rayleigh scattering favours shorter wavelengths, Mie scattering acts with larger particles and the Tyndall effect reveals colloids. C. Temperature describes thermal state and is not heat; heat is energy transferred because of a temperature difference. Specific heat compares energy required for a temperature rise, while latent heat is absorbed or released during a phase change without a temperature change. D. Reflection obeys equality of incidence and reflection angles; plane, concave and convex mirrors form images according to geometry and object position. Mirror action is reflection, so it must not be explained through the refraction mechanism used by lenses.

Answer: C. Explanation: Temperature describes thermal state and is not heat; heat is energy transferred because of a temperature difference. Specific heat compares energy required for a temperature rise, while latent heat is absorbed or released during a phase change without a temperature change. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q28. Which option avoids the standard UPSC close-option trap about Heat temperature specific and latent heat?

A. Reflection obeys equality of incidence and reflection angles; plane, concave and convex mirrors form images according to geometry and object position. Mirror action is reflection, so it must not be explained through the refraction mechanism used by lenses. B. Refraction follows a change in light speed across a boundary while frequency remains unchanged; total internal reflection requires travel from denser to rarer medium beyond the critical angle and underpins optical fibres. Convex and concave lenses converge and diverge respectively, with myopia corrected by a concave lens, hypermetropia by a convex lens, presbyopia commonly by bifocals and astigmatism by cylindrical lenses. Dispersion separates colours, Rayleigh scattering favours shorter wavelengths, Mie scattering acts with larger particles and the Tyndall effect reveals colloids. C. Electric current is charge flow per unit time, potential difference is work per unit charge and resistance opposes current. Ohm's law V = IR applies under stated physical conditions to ohmic conductors, so it is not a universal description of every material or device. D. Temperature describes thermal state and is not heat; heat is energy transferred because of a temperature difference. Specific heat compares energy required for a temperature rise, while latent heat is absorbed or released during a phase change without a temperature change.

Answer: D. Explanation: Temperature describes thermal state and is not heat; heat is energy transferred because of a temperature difference. Specific heat compares energy required for a temperature rise, while latent heat is absorbed or released during a phase change without a temperature change. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q29. Which statement correctly identifies Thermodynamics and heat transfer?

A. Conduction, convection and radiation are distinct heat-transfer modes, and radiation needs no material medium. The first law applies energy conservation to thermal systems; the second law fixes spontaneous heat flow from hotter to colder bodies and rules out a perfectly efficient heat engine. An ideal black body absorbs and emits across wavelengths, and hotter bodies shift peak emission toward shorter wavelengths. B. In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence. C. A wave transfers energy without net transport of matter, with speed equal to frequency multiplied by wavelength. Sound is mechanical and travels fastest in solids and slowest in gases; frequency primarily sets pitch, amplitude affects loudness, Doppler shift follows relative motion, and resonance amplifies forced oscillation near natural frequency. Infrasound is below 20 Hz, audible sound spans about 20 Hz to 20 kHz and ultrasound is above 20 kHz; ultrasound must not be confused with supersonic motion, while echo and reverberation differ by time separation. D. Reflection obeys equality of incidence and reflection angles; plane, concave and convex mirrors form images according to geometry and object position. Mirror action is reflection, so it must not be explained through the refraction mechanism used by lenses.

Answer: A. Explanation: Conduction, convection and radiation are distinct heat-transfer modes, and radiation needs no material medium. The first law applies energy conservation to thermal systems; the second law

fixes spontaneous heat flow from hotter to colder bodies and rules out a perfectly efficient heat engine. An ideal black body absorbs and emits across wavelengths, and hotter bodies shift peak emission toward shorter wavelengths. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q30. Which option preserves the technical boundary of Thermodynamics and heat transfer?

A. Reflection obeys equality of incidence and reflection angles; plane, concave and convex mirrors form images according to geometry and object position. Mirror action is reflection, so it must not be explained through the refraction mechanism used by lenses. B. Conduction, convection and radiation are distinct heat-transfer modes, and radiation needs no material medium. The first law applies energy conservation to thermal systems; the second law fixes spontaneous heat flow from hotter to colder bodies and rules out a perfectly efficient heat engine. An ideal black body absorbs and emits across wavelengths, and hotter bodies shift peak emission toward shorter wavelengths. C. In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence. D. Refraction follows a change in light speed across a boundary while frequency remains unchanged; total internal reflection requires travel from denser to rarer medium beyond the critical angle and underpins optical fibres. Convex and concave lenses converge and diverge respectively, with myopia corrected by a concave lens, hypermetropia by a convex lens, presbyopia commonly by bifocals and astigmatism by cylindrical lenses. Dispersion separates colours, Rayleigh scattering favours shorter wavelengths, Mie scattering acts with larger particles and the Tyndall effect reveals colloids.

Answer: B. Explanation: Conduction, convection and radiation are distinct heat-transfer modes, and radiation needs no material medium. The first law applies energy conservation to thermal systems; the second law fixes spontaneous heat flow from hotter to colder bodies and rules out a perfectly efficient heat engine. An ideal black body absorbs and emits across wavelengths, and hotter bodies shift peak emission toward shorter wavelengths. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q31. Which statement uses Thermodynamics and heat transfer without changing its institution, unit or status?

A. Electric current is charge flow per unit time, potential difference is work per unit charge and resistance opposes current. Ohm's law V = IR applies under stated physical conditions to ohmic conductors, so it is not a universal description of every material or device. B. Reflection obeys equality of incidence and reflection angles; plane, concave and convex mirrors form images according to geometry and object position. Mirror action is reflection, so it must not be explained through the refraction mechanism used by lenses. C. Conduction, convection and radiation are distinct heat-transfer modes, and radiation needs no material medium. The first law applies energy conservation to thermal systems; the second law fixes spontaneous heat flow from hotter to colder bodies and rules out a perfectly efficient heat engine. An ideal black body absorbs and emits across wavelengths, and hotter bodies shift peak emission toward shorter wavelengths. D. Refraction follows a change in light speed across a boundary while frequency remains unchanged; total internal reflection requires travel from denser to rarer medium beyond the critical angle and underpins optical fibres. Convex and concave lenses converge and diverge respectively, with myopia corrected by a concave lens, hypermetropia by a convex lens, presbyopia commonly by bifocals and astigmatism by cylindrical lenses. Dispersion separates colours, Rayleigh scattering favours shorter wavelengths, Mie scattering acts with larger particles and the Tyndall effect reveals colloids.

Answer: C. Explanation: Conduction, convection and radiation are distinct heat-transfer modes, and radiation needs no material medium. The first law applies energy conservation to thermal systems; the second law fixes spontaneous heat flow from hotter to colder bodies and rules out a perfectly efficient heat engine. An ideal black body absorbs and emits across wavelengths, and hotter bodies shift peak emission toward shorter wavelengths. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q32. Which option avoids the standard UPSC close-option trap about Thermodynamics and heat transfer?

A. Electric current is charge flow per unit time, potential difference is work per unit charge and resistance opposes current. Ohm's law V = IR applies under stated physical conditions to ohmic conductors, so it is not a universal description of every material or device. B. Series branches share current and parallel branches share voltage; domestic loads are connected in parallel. Electrical power can be written VI under the relevant circuit conditions, transmission loss rises with I squared R, and fuse or MCB protection and earthing perform different safety functions. Below a critical temperature, superconductors show zero electrical resistance and the Meissner effect, supporting specialised magnets rather than ordinary room-temperature wiring claims. C. Refraction follows a change in light speed across a boundary while frequency remains unchanged; total internal reflection requires travel from denser to rarer medium beyond the critical angle and underpins optical fibres. Convex and concave lenses converge and diverge respectively, with myopia corrected by a concave lens, hypermetropia by a convex lens, presbyopia commonly by bifocals and astigmatism by cylindrical lenses. Dispersion separates colours, Rayleigh scattering favours shorter wavelengths, Mie scattering acts with larger particles and the Tyndall effect reveals colloids. D. Conduction, convection and radiation are distinct heat-transfer modes, and radiation needs no material medium. The first law applies energy conservation to thermal systems; the second law fixes spontaneous heat flow from hotter to colder bodies and rules out a perfectly efficient heat engine. An ideal black body absorbs and emits across wavelengths, and hotter bodies shift peak emission toward shorter wavelengths.

Answer: D. Explanation: Conduction, convection and radiation are distinct heat-transfer modes, and radiation needs no material medium. The first law applies energy conservation to thermal systems; the second law fixes spontaneous heat flow from hotter to colder bodies and rules out a perfectly efficient heat engine. An ideal black body absorbs and emits across wavelengths, and hotter bodies shift peak emission toward shorter wavelengths. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q33. Which statement correctly identifies Wave relation and sound?

A. A wave transfers energy without net transport of matter, with speed equal to frequency multiplied by wavelength. Sound is mechanical and travels fastest in solids and slowest in gases; frequency primarily sets pitch, amplitude affects loudness, Doppler shift follows relative motion, and resonance amplifies forced oscillation near natural frequency. Infrasound is below 20 Hz, audible sound spans about 20 Hz to 20 kHz and ultrasound is above 20 kHz; ultrasound must not be confused with supersonic motion, while echo and reverberation differ by time separation. B. Refraction follows a change in light speed across a boundary while frequency remains unchanged; total internal reflection requires travel from denser to rarer medium beyond the critical angle and underpins optical fibres. Convex and concave lenses converge and diverge respectively, with myopia corrected by a concave lens, hypermetropia by a convex lens, presbyopia commonly by bifocals and astigmatism by cylindrical lenses. Dispersion separates colours, Rayleigh scattering favours shorter wavelengths, Mie scattering acts with larger particles and the Tyndall effect reveals colloids. C. Reflection obeys equality of incidence and reflection angles; plane, concave and convex mirrors form images according to geometry and object position. Mirror action is reflection, so it must not be explained through the refraction mechanism used by lenses. D. In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence.

Answer: A. Explanation: A wave transfers energy without net transport of matter, with speed equal to frequency multiplied by wavelength. Sound is mechanical and travels fastest in solids and slowest in gases; frequency primarily sets pitch, amplitude affects loudness, Doppler shift follows relative motion, and resonance amplifies forced oscillation near natural frequency. Infrasound is below 20 Hz, audible sound spans about 20 Hz to 20 kHz and ultrasound is above 20 kHz; ultrasound must not be confused with supersonic motion, while echo and reverberation differ by time separation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q34. Which option preserves the technical boundary of Wave relation and sound?

A. Refraction follows a change in light speed across a boundary while frequency remains unchanged; total internal reflection requires travel from denser to rarer medium beyond the critical angle and underpins optical fibres. Convex and concave lenses converge and diverge respectively, with myopia corrected by a concave lens, hypermetropia by a convex lens, presbyopia commonly by bifocals and astigmatism by cylindrical lenses. Dispersion separates colours, Rayleigh scattering favours shorter wavelengths, Mie scattering acts with larger particles and the Tyndall effect reveals colloids. B. A wave transfers energy without net transport of matter, with speed equal to frequency multiplied by wavelength. Sound is mechanical and travels fastest in solids and slowest in gases; frequency primarily sets pitch, amplitude affects loudness, Doppler shift follows relative motion, and resonance amplifies forced oscillation near natural frequency. Infrasound is below 20 Hz, audible sound spans about 20 Hz to 20 kHz and ultrasound is above 20 kHz; ultrasound must not be confused with supersonic motion, while echo and reverberation differ by time separation. C. Reflection obeys equality of incidence and reflection angles; plane, concave and convex mirrors form images according to geometry and object position. Mirror action is reflection, so it must not be explained through the refraction mechanism used by lenses. D. Electric current is charge flow per unit time, potential difference is work per unit charge and resistance opposes current. Ohm's law V = IR applies under stated physical conditions to ohmic conductors, so it is not a universal description of every material or device.

Answer: B. Explanation: A wave transfers energy without net transport of matter, with speed equal to frequency multiplied by wavelength. Sound is mechanical and travels fastest in solids and slowest in gases; frequency primarily sets pitch, amplitude affects loudness, Doppler shift follows relative motion, and resonance amplifies forced oscillation near natural frequency. Infrasound is below 20 Hz, audible sound spans about 20 Hz to 20 kHz and ultrasound is above 20 kHz; ultrasound must not be confused with supersonic motion, while echo and reverberation differ by time separation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q35. Which statement uses Wave relation and sound without changing its institution, unit or status?

A. Series branches share current and parallel branches share voltage; domestic loads are connected in parallel. Electrical power can be written VI under the relevant circuit conditions, transmission loss rises with I squared R, and fuse or MCB protection and earthing perform different safety functions. Below a critical temperature, superconductors show zero electrical resistance and the Meissner effect, supporting specialised magnets rather than ordinary room-temperature wiring claims. B. Electric current is charge flow per unit time, potential difference is work per unit charge and resistance opposes current. Ohm's law V = IR applies under stated physical conditions to ohmic conductors, so it is not a universal description of every material or device. C. A wave transfers energy without net transport of matter, with speed equal to frequency multiplied by wavelength. Sound is mechanical and travels fastest in solids and slowest in gases; frequency primarily sets pitch, amplitude affects loudness, Doppler shift follows relative motion, and resonance amplifies forced oscillation near natural frequency. Infrasound is below 20 Hz, audible sound spans about 20 Hz to 20 kHz and ultrasound is above 20 kHz; ultrasound must not be confused with supersonic motion, while echo and reverberation differ by time separation. D. Refraction follows a change in light speed across a boundary while frequency remains unchanged; total internal reflection requires travel from denser to rarer medium beyond the critical angle and underpins optical fibres. Convex and concave lenses converge and diverge respectively, with myopia corrected by a concave lens, hypermetropia by a convex lens, presbyopia commonly by bifocals and astigmatism by cylindrical lenses. Dispersion separates colours, Rayleigh scattering favours shorter wavelengths, Mie scattering acts with larger particles and the Tyndall effect reveals colloids.

Answer: C. Explanation: A wave transfers energy without net transport of matter, with speed equal to frequency multiplied by wavelength. Sound is mechanical and travels fastest in solids and slowest in gases; frequency primarily sets pitch, amplitude affects loudness, Doppler shift follows relative motion, and resonance amplifies forced oscillation near natural frequency. Infrasound is below 20 Hz, audible sound spans about 20 Hz to 20 kHz and ultrasound is above 20 kHz; ultrasound must not be confused with supersonic motion, while echo and reverberation differ by time separation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q36. Which option avoids the standard UPSC close-option trap about Wave relation and sound?

A. Electric current is charge flow per unit time, potential difference is work per unit charge and resistance opposes current. Ohm's law V = IR applies under stated physical conditions to ohmic conductors, so it is not a universal description of every material or device. B. Changing magnetic flux induces an emf; generators convert mechanical to electrical energy, motors convert electrical to mechanical energy, and transformers use mutual induction to change AC voltage. A steady DC supply does not provide the changing flux needed for ordinary transformer action. C. Series branches share current and parallel branches share voltage; domestic loads are connected in parallel. Electrical power can be written VI under the relevant circuit conditions, transmission loss rises with I squared R, and fuse or MCB protection and earthing perform different safety functions. Below a critical temperature, superconductors show zero electrical resistance and the Meissner effect, supporting specialised magnets rather than ordinary room-temperature wiring claims. D. A wave transfers energy without net transport of matter, with speed equal to frequency multiplied by wavelength. Sound is mechanical and travels fastest in solids and slowest in gases; frequency primarily sets pitch, amplitude affects loudness, Doppler shift follows relative motion, and resonance amplifies forced oscillation near natural frequency. Infrasound is below 20 Hz, audible sound spans about 20 Hz to 20 kHz and ultrasound is above 20 kHz; ultrasound must not be confused with supersonic motion, while echo and reverberation differ by time separation.

Answer: D. Explanation: A wave transfers energy without net transport of matter, with speed equal to frequency multiplied by wavelength. Sound is mechanical and travels fastest in solids and slowest in gases; frequency primarily sets pitch, amplitude affects loudness, Doppler shift follows relative motion, and resonance amplifies forced oscillation near natural frequency. Infrasound is below 20 Hz, audible sound spans about 20 Hz to 20 kHz and ultrasound is above 20 kHz; ultrasound must not be confused with supersonic motion, while echo and reverberation differ by time separation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q37. Which statement correctly identifies Electromagnetic spectrum and communication boundary?

A. In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence. B. Refraction follows a change in light speed across a boundary while frequency remains unchanged; total internal reflection requires travel from denser to rarer medium beyond the critical angle and underpins optical fibres. Convex and concave lenses converge and diverge respectively, with myopia corrected by a concave lens, hypermetropia by a convex lens, presbyopia commonly by bifocals and astigmatism by cylindrical lenses. Dispersion separates colours, Rayleigh scattering favours shorter wavelengths, Mie scattering acts with larger particles and the Tyndall effect reveals colloids. C. Electric current is charge flow per unit time, potential difference is work per unit charge and resistance opposes current. Ohm's law V = IR applies under stated physical conditions to ohmic conductors, so it is not a universal description of every material or device. D. Reflection obeys equality of incidence and reflection angles; plane, concave and convex mirrors form images according to geometry and object position. Mirror action is reflection, so it must not be explained through the refraction mechanism used by lenses.

Answer: A. Explanation: In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q38. Which option preserves the technical boundary of Electromagnetic spectrum and communication boundary?

A. Series branches share current and parallel branches share voltage; domestic loads are connected in parallel. Electrical power can be written VI under the relevant circuit conditions, transmission loss rises with I squared R, and fuse or MCB protection and earthing perform different safety functions. Below a critical temperature, superconductors show zero electrical resistance and the Meissner effect, supporting specialised magnets rather than ordinary room-temperature wiring claims. B. In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence. C. Electric current is charge flow per unit time, potential difference is work per unit charge and resistance opposes current. Ohm's law V = IR applies under stated physical conditions to ohmic conductors, so it is not a universal description of every material or device. D. Refraction follows a change in light speed across a boundary while frequency remains unchanged; total internal reflection requires travel from denser to rarer medium beyond the critical angle and underpins optical fibres. Convex and concave lenses converge and diverge respectively, with myopia corrected by a concave lens, hypermetropia by a convex lens, presbyopia commonly by bifocals and astigmatism by cylindrical lenses. Dispersion separates colours, Rayleigh scattering favours shorter wavelengths, Mie scattering acts with larger particles and the Tyndall effect reveals colloids.

Answer: B. Explanation: In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q39. Which statement uses Electromagnetic spectrum and communication boundary without changing its institution, unit or status?

A. Series branches share current and parallel branches share voltage; domestic loads are connected in parallel. Electrical power can be written VI under the relevant circuit conditions, transmission loss rises with I squared R, and fuse or MCB protection and earthing perform different safety functions. Below a critical temperature, superconductors show zero electrical resistance and the Meissner effect, supporting specialised magnets rather than ordinary room-temperature wiring claims. B. Changing magnetic flux induces an emf; generators convert mechanical to electrical energy, motors convert electrical to mechanical energy, and transformers use mutual induction to change AC voltage. A steady DC supply does not provide the changing flux needed for ordinary transformer action. C. In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence. D. Electric current is charge flow per unit time, potential difference is work per unit charge and resistance opposes current. Ohm's law V = IR applies under stated physical conditions to ohmic conductors, so it is not a universal description of every material or device.

Answer: C. Explanation: In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q40. Which option avoids the standard UPSC close-option trap about Electromagnetic spectrum and communication boundary?

A. Changing magnetic flux induces an emf; generators convert mechanical to electrical energy, motors convert electrical to mechanical energy, and transformers use mutual induction to change AC voltage. A steady DC supply does not provide the changing flux needed for ordinary transformer action. B. Series branches share current and parallel branches share voltage; domestic loads are connected in parallel. Electrical power can be written VI under the relevant circuit conditions, transmission loss rises with I squared R, and fuse or MCB protection and earthing perform different safety functions. Below a critical temperature, superconductors show zero electrical resistance and the Meissner effect, supporting specialised magnets rather than ordinary room-temperature wiring claims. C. Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to brightness, treated as interchangeable or extended to every particle. D. In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence.

Answer: D. Explanation: In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q41. Which statement correctly identifies Reflection mirrors and image logic?

A. Reflection obeys equality of incidence and reflection angles; plane, concave and convex mirrors form images according to geometry and object position. Mirror action is reflection, so it must not be explained through the refraction mechanism used by lenses. B. Series branches share current and parallel branches share voltage; domestic loads are connected in parallel. Electrical power can be written VI under the relevant circuit conditions, transmission loss rises with I squared R, and fuse or MCB protection and earthing perform different safety functions. Below a critical temperature, superconductors show zero electrical resistance and the Meissner effect, supporting specialised magnets rather than ordinary room-temperature wiring claims. C. Refraction follows a change in light speed across a boundary while frequency remains unchanged; total internal reflection requires travel from denser to rarer medium beyond the critical angle and underpins optical fibres. Convex and concave lenses converge and diverge respectively, with myopia corrected by a concave lens, hypermetropia by a convex lens, presbyopia commonly by bifocals and astigmatism by cylindrical lenses. Dispersion separates colours, Rayleigh scattering favours shorter wavelengths, Mie scattering acts with larger particles and the Tyndall effect reveals colloids. D. Electric current is charge flow per unit time, potential difference is work per unit charge and resistance opposes current. Ohm's law $V = IR$ applies under stated physical conditions to ohmic conductors, so it is not a universal description of every material or device.

Answer: A. Explanation: Reflection obeys equality of incidence and reflection angles; plane, concave and convex mirrors form images according to geometry and object position. Mirror action is reflection, so it must not be explained through the refraction mechanism used by lenses. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q42. Which option preserves the technical boundary of Reflection mirrors and image logic?

A. Series branches share current and parallel branches share voltage; domestic loads are connected in parallel. Electrical power can be written VI under the relevant circuit conditions, transmission loss rises with I squared R, and fuse or MCB protection and earthing perform different safety functions. Below a critical temperature, superconductors show zero electrical resistance and the Meissner effect, supporting specialised magnets rather than ordinary room-temperature wiring claims. B. Reflection obeys equality of incidence and reflection angles; plane, concave and convex mirrors form images according to geometry and object position. Mirror action is reflection, so it must not be explained through the refraction mechanism used by lenses. C. Changing magnetic flux induces an emf; generators convert mechanical to electrical energy, motors convert electrical to mechanical energy, and transformers use mutual induction to change AC voltage. A steady DC supply does not provide the changing flux needed for ordinary transformer action. D. Electric current is charge flow per unit time, potential difference is work per unit charge and resistance opposes current. Ohm's law V = IR applies under stated physical conditions to ohmic conductors, so it is not a universal description of every material or device.

Answer: B. Explanation: Reflection obeys equality of incidence and reflection angles; plane, concave and convex mirrors form images according to geometry and object position. Mirror action is reflection, so it must not be explained through the refraction mechanism used by lenses. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q43. Which statement uses Reflection mirrors and image logic without changing its institution, unit or status?

A. Changing magnetic flux induces an emf; generators convert mechanical to electrical energy, motors convert electrical to mechanical energy, and transformers use mutual induction to change AC voltage. A steady DC supply does not provide the changing flux needed for ordinary transformer action. B. Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to brightness, treated as interchangeable or extended to every particle. C. Reflection obeys equality of incidence and reflection angles; plane, concave and convex mirrors form images according to geometry and object position. Mirror action is reflection, so it must not be explained through the refraction mechanism used by lenses. D. Series branches share current and parallel branches share voltage; domestic loads are connected in parallel. Electrical power can be written VI under the relevant circuit conditions, transmission loss rises with I squared R, and fuse or MCB protection and earthing perform different safety functions. Below a critical temperature, superconductors show zero electrical resistance and the Meissner effect, supporting specialised magnets rather than ordinary room-temperature wiring claims.

Answer: C. Explanation: Reflection obeys equality of incidence and reflection angles; plane, concave and convex mirrors form images according to geometry and object position. Mirror action is reflection, so it must not be explained through the refraction mechanism used by lenses. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q44. Which option avoids the standard UPSC close-option trap about Reflection mirrors and image logic?

A. Changing magnetic flux induces an emf; generators convert mechanical to electrical energy, motors convert electrical to mechanical energy, and transformers use mutual induction to change AC voltage. A steady DC supply does not provide the changing flux needed for ordinary transformer action. B. Alpha radiation is a helium nucleus, beta commonly involves an electron or positron, and gamma is electromagnetic radiation; alpha is strongly ionising but weakly penetrating relative to gamma, whose penetration is much greater. Half-life is a nuclear decay property unaffected by ordinary temperature or pressure and supports radioisotope dating such as carbon-14, while fission splits heavy nuclei and fusion joins light nuclei. C. Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to brightness, treated as interchangeable or extended to every particle. D. Reflection obeys equality of incidence and reflection angles; plane, concave and convex mirrors form images according to geometry and object position. Mirror action is reflection, so it must not be explained through the refraction mechanism used by lenses.

Answer: D. Explanation: Reflection obeys equality of incidence and reflection angles; plane, concave and convex mirrors form images according to geometry and object position. Mirror action is reflection, so it must not be explained through the refraction mechanism used by lenses. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q45. Which statement correctly identifies Refraction total internal reflection and lenses?

A. Refraction follows a change in light speed across a boundary while frequency remains unchanged; total internal reflection requires travel from denser to rarer medium beyond the critical angle and underpins optical fibres. Convex and concave lenses converge and diverge respectively, with myopia corrected by a concave lens, hypermetropia by a convex lens, presbyopia commonly by bifocals and astigmatism by cylindrical lenses. Dispersion separates colours, Rayleigh scattering favours shorter wavelengths, Mie scattering acts with larger particles and the Tyndall effect reveals colloids. B. Electric current is charge flow per unit time, potential difference is work per unit charge and resistance opposes current. Ohm's law V = IR applies under stated physical conditions to ohmic conductors, so it is not a universal description of every material or device. C. Changing magnetic flux induces an emf; generators convert mechanical to electrical energy, motors convert electrical to mechanical energy, and transformers use mutual induction to change AC voltage. A steady DC supply does not provide the changing flux needed for ordinary transformer action. D. Series branches share current and parallel branches share voltage; domestic loads are connected in parallel. Electrical power can be written VI under the relevant circuit conditions, transmission loss rises with I squared R, and fuse or MCB protection and earthing perform different safety functions. Below a critical temperature, superconductors show zero electrical resistance and the Meissner effect, supporting specialised magnets rather than ordinary room-temperature wiring claims.

Answer: A. Explanation: Refraction follows a change in light speed across a boundary while frequency remains unchanged; total internal reflection requires travel from denser to rarer medium beyond the critical angle and underpins optical fibres. Convex and concave lenses converge and diverge respectively, with myopia corrected by a concave lens, hypermetropia by a convex lens, presbyopia commonly by bifocals and astigmatism by cylindrical lenses. Dispersion separates colours, Rayleigh scattering favours shorter wavelengths, Mie scattering acts with larger particles and the Tyndall effect reveals colloids. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q46. Which option preserves the technical boundary of Refraction total internal reflection and lenses?

A. Series branches share current and parallel branches share voltage; domestic loads are connected in parallel. Electrical power can be written VI under the relevant circuit conditions, transmission loss rises with I squared R, and fuse or MCB protection and earthing perform different safety functions. Below a critical temperature, superconductors show zero electrical resistance and the Meissner effect, supporting specialised magnets rather than ordinary room-temperature wiring claims. B. Refraction follows a change in light speed across a boundary while frequency remains unchanged; total internal reflection requires travel from denser to rarer medium beyond the critical angle and underpins optical fibres. Convex and concave lenses converge and diverge respectively, with myopia corrected by a concave lens, hypermetropia by a convex lens, presbyopia commonly by bifocals and astigmatism by cylindrical lenses. Dispersion separates colours, Rayleigh scattering favours shorter wavelengths, Mie scattering acts with larger particles and the Tyndall effect reveals colloids. C. Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to brightness, treated as interchangeable or extended to every particle. D. Changing magnetic flux induces an emf; generators convert mechanical to electrical energy, motors convert electrical to mechanical energy, and transformers use mutual induction to change AC voltage. A steady DC supply does not provide the changing flux needed for ordinary transformer action.

Answer: B. Explanation: Refraction follows a change in light speed across a boundary while frequency remains unchanged; total internal reflection requires travel from denser to rarer medium beyond the critical

angle and underpins optical fibres. Convex and concave lenses converge and diverge respectively, with myopia corrected by a concave lens, hypermetropia by a convex lens, presbyopia commonly by bifocals and astigmatism by cylindrical lenses. Dispersion separates colours, Rayleigh scattering favours shorter wavelengths, Mie scattering acts with larger particles and the Tyndall effect reveals colloids. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

Q47. Which statement uses Refraction total internal reflection and lenses without changing its institution, unit or status?

A. Changing magnetic flux induces an emf; generators convert mechanical to electrical energy, motors convert electrical to mechanical energy, and transformers use mutual induction to change AC voltage. A steady DC supply does not provide the changing flux needed for ordinary transformer action. B. Alpha radiation is a helium nucleus, beta commonly involves an electron or positron, and gamma is electromagnetic radiation; alpha is strongly ionising but weakly penetrating relative to gamma, whose penetration is much greater. Half-life is a nuclear decay property unaffected by ordinary temperature or pressure and supports radioisotope dating such as carbon-14, while fission splits heavy nuclei and fusion joins light nuclei. C. Refraction follows a change in light speed across a boundary while frequency remains unchanged; total internal reflection requires travel from denser to rarer medium beyond the critical angle and underpins optical fibres. Convex and concave lenses converge and diverge respectively, with myopia corrected by a concave lens, hypermetropia by a convex lens, presbyopia commonly by bifocals and astigmatism by cylindrical lenses. Dispersion separates colours, Rayleigh scattering favours shorter wavelengths, Mie scattering acts with larger particles and the Tyndall effect reveals colloids. D. Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to brightness, treated as interchangeable or extended to every particle.

Answer: C. Explanation: Refraction follows a change in light speed across a boundary while frequency remains unchanged; total internal reflection requires travel from denser to rarer medium beyond the critical angle and underpins optical fibres. Convex and concave lenses converge and diverge respectively, with myopia corrected by a concave lens, hypermetropia by a convex lens, presbyopia commonly by bifocals and astigmatism by cylindrical lenses. Dispersion separates colours, Rayleigh scattering favours shorter wavelengths, Mie scattering acts with larger particles and the Tyndall effect reveals colloids. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q48. Which option avoids the standard UPSC close-option trap about Refraction total internal reflection and lenses?

A. Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena. B. Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to brightness, treated as interchangeable or extended to every particle. C. Alpha radiation is a helium nucleus, beta commonly involves an electron or positron, and gamma is electromagnetic radiation; alpha is strongly ionising but weakly penetrating relative to gamma, whose penetration is much greater. Half-life is a nuclear decay property unaffected by ordinary temperature or pressure and supports radioisotope dating such as carbon-14, while fission splits heavy nuclei and fusion joins light nuclei. D. Refraction follows a change in light speed across a boundary while frequency remains unchanged; total internal reflection requires travel from denser to rarer medium beyond the critical angle and underpins optical fibres. Convex and concave lenses converge and diverge respectively, with myopia corrected by a concave lens, hypermetropia by a convex lens, presbyopia commonly by bifocals and astigmatism by cylindrical lenses. Dispersion separates colours, Rayleigh scattering favours shorter wavelengths, Mie scattering acts with larger particles and the Tyndall effect reveals colloids.

**Answer: D. Explanation: Refraction follows a change in light speed across a boundary while frequency remains unchanged; total internal reflection requires travel from denser to rarer medium beyond the critical

angle and underpins optical fibres. Convex and concave lenses converge and diverge respectively, with myopia corrected by a concave lens, hypermetropia by a convex lens, presbyopia commonly by bifocals and astigmatism by cylindrical lenses. Dispersion separates colours, Rayleigh scattering favours shorter wavelengths, Mie scattering acts with larger particles and the Tyndall effect reveals colloids. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q49. Which statement correctly identifies Current voltage resistance and Ohm boundary?

A. Electric current is charge flow per unit time, potential difference is work per unit charge and resistance opposes current. Ohm's law V = IR applies under stated physical conditions to ohmic conductors, so it is not a universal description of every material or device. B. Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to brightness, treated as interchangeable or extended to every particle. C. Series branches share current and parallel branches share voltage; domestic loads are connected in parallel. Electrical power can be written VI under the relevant circuit conditions, transmission loss rises with I squared R, and fuse or MCB protection and earthing perform different safety functions. Below a critical temperature, superconductors show zero electrical resistance and the Meissner effect, supporting specialised magnets rather than ordinary room-temperature wiring claims. D. Changing magnetic flux induces an emf; generators convert mechanical to electrical energy, motors convert electrical to mechanical energy, and transformers use mutual induction to change AC voltage. A steady DC supply does not provide the changing flux needed for ordinary transformer action.

Answer: A. Explanation: Electric current is charge flow per unit time, potential difference is work per unit charge and resistance opposes current. Ohm's law V = IR applies under stated physical conditions to ohmic conductors, so it is not a universal description of every material or device. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q50. Which option preserves the technical boundary of Current voltage resistance and Ohm boundary?

A. Alpha radiation is a helium nucleus, beta commonly involves an electron or positron, and gamma is electromagnetic radiation; alpha is strongly ionising but weakly penetrating relative to gamma, whose penetration is much greater. Half-life is a nuclear decay property unaffected by ordinary temperature or pressure and supports radioisotope dating such as carbon-14, while fission splits heavy nuclei and fusion joins light nuclei. B. Electric current is charge flow per unit time, potential difference is work per unit charge and resistance opposes current. Ohm's law V = IR applies under stated physical conditions to ohmic conductors, so it is not a universal description of every material or device. C. Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to brightness, treated as interchangeable or extended to every particle. D. Changing magnetic flux induces an emf; generators convert mechanical to electrical energy, motors convert electrical to mechanical energy, and transformers use mutual induction to change AC voltage. A steady DC supply does not provide the changing flux needed for ordinary transformer action.

Answer: B. Explanation: Electric current is charge flow per unit time, potential difference is work per unit charge and resistance opposes current. Ohm's law V = IR applies under stated physical conditions to ohmic conductors, so it is not a universal description of every material or device. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q51. Which statement uses Current voltage resistance and Ohm boundary without changing its institution, unit or status?

A. Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena. B. Alpha radiation is a helium nucleus, beta commonly involves an electron or positron, and gamma is electromagnetic radiation; alpha is strongly ionising but weakly penetrating relative to gamma, whose penetration is much greater. Half-life is a nuclear decay property unaffected by ordinary temperature or pressure and supports radioisotope dating such as carbon-14, while fission splits heavy nuclei and fusion joins light nuclei. C. Electric current is charge flow per unit time, potential difference is work per unit charge and resistance opposes current. Ohm's law V = IR applies under stated physical conditions to ohmic conductors, so it is not a universal description of every material or device. D. Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to brightness, treated as interchangeable or extended to every particle.

Answer: C. Explanation: Electric current is charge flow per unit time, potential difference is work per unit charge and resistance opposes current. Ohm's law V = IR applies under stated physical conditions to ohmic conductors, so it is not a universal description of every material or device. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q52. Which option avoids the standard UPSC close-option trap about Current voltage resistance and Ohm boundary?

A. Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability. B. Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena. C. Alpha radiation is a helium nucleus, beta commonly involves an electron or positron, and gamma is electromagnetic radiation; alpha is strongly ionising but weakly penetrating relative to gamma, whose penetration is much greater. Half-life is a nuclear decay property unaffected by ordinary temperature or pressure and supports radioisotope dating such as carbon-14, while fission splits heavy nuclei and fusion joins light nuclei. D. Electric current is charge flow per unit time, potential difference is work per unit charge and resistance opposes current. Ohm's law V = IR applies under stated physical conditions to ohmic conductors, so it is not a universal description of every material or device.

Answer: D. Explanation: Electric current is charge flow per unit time, potential difference is work per unit charge and resistance opposes current. Ohm's law V = IR applies under stated physical conditions to ohmic conductors, so it is not a universal description of every material or device. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q53. Which statement correctly identifies Circuits electrical power and safety?

A. Series branches share current and parallel branches share voltage; domestic loads are connected in parallel. Electrical power can be written VI under the relevant circuit conditions, transmission loss rises with I squared R, and fuse or MCB protection and earthing perform different safety functions. Below a critical temperature, superconductors show zero electrical resistance and the Meissner effect, supporting specialised magnets rather than ordinary room-temperature wiring claims. B. Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to brightness, treated as interchangeable or extended to every particle. C. Alpha radiation is a helium nucleus, beta commonly involves an electron or positron, and gamma is electromagnetic radiation; alpha is strongly ionising but weakly penetrating relative to gamma, whose penetration is much greater. Half-life is a nuclear decay property unaffected by ordinary temperature or pressure and supports radioisotope dating such as carbon-14, while fission splits heavy nuclei and fusion joins light nuclei. D. Changing magnetic flux induces an emf; generators convert mechanical to electrical energy, motors convert electrical to mechanical energy, and transformers use mutual induction to change AC voltage. A steady DC supply does not provide the changing flux needed for ordinary transformer action.

Answer: A. Explanation: Series branches share current and parallel branches share voltage; domestic loads are connected in parallel. Electrical power can be written VI under the relevant circuit conditions, transmission loss rises with I squared R, and fuse or MCB protection and earthing perform different safety functions. Below a critical temperature, superconductors show zero electrical resistance and the Meissner effect, supporting specialised magnets rather than ordinary room-temperature wiring claims. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q54. Which option preserves the technical boundary of Circuits electrical power and safety?

A. Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena. B. Series branches share current and parallel branches share voltage; domestic loads are connected in parallel. Electrical power can be written VI under the relevant circuit conditions, transmission loss rises with I squared R, and fuse or MCB protection and earthing perform different safety functions. Below a critical temperature, superconductors show zero electrical resistance and the Meissner effect, supporting specialised magnets rather than ordinary room-temperature wiring claims. C. Alpha radiation is a helium nucleus, beta commonly involves an electron or positron, and gamma is electromagnetic radiation; alpha is strongly ionising but weakly penetrating relative to gamma, whose penetration is much greater. Half-life is a nuclear decay property unaffected by ordinary temperature or pressure and supports radioisotope dating such as carbon-14, while fission splits heavy nuclei and fusion joins light nuclei. D. Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to brightness, treated as interchangeable or extended to every particle.

Answer: B. Explanation: Series branches share current and parallel branches share voltage; domestic loads are connected in parallel. Electrical power can be written VI under the relevant circuit conditions, transmission loss rises with I squared R, and fuse or MCB protection and earthing perform different safety functions. Below a critical temperature, superconductors show zero electrical resistance and the Meissner effect, supporting specialised magnets rather than ordinary room-temperature wiring claims. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q55. Which statement uses Circuits electrical power and safety without changing its institution, unit or status?

A. Alpha radiation is a helium nucleus, beta commonly involves an electron or positron, and gamma is electromagnetic radiation; alpha is strongly ionising but weakly penetrating relative to gamma, whose penetration is much greater. Half-life is a nuclear decay property unaffected by ordinary temperature or pressure and supports radioisotope dating such as carbon-14, while fission splits heavy nuclei and fusion joins light nuclei. B. Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena. C. Series branches share current and parallel branches share voltage; domestic loads are connected in parallel. Electrical power can be written VI under the relevant circuit conditions, transmission loss rises with I squared R, and fuse or MCB protection and earthing perform different safety functions. Below a critical temperature, superconductors show zero electrical resistance and the Meissner effect, supporting specialised magnets rather than ordinary room-temperature wiring claims. D. Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability.

Answer: C. Explanation: Series branches share current and parallel branches share voltage; domestic loads are connected in parallel. Electrical power can be written VI under the relevant circuit conditions, transmission loss rises with I squared R, and fuse or MCB protection and earthing perform different safety functions. Below a critical temperature, superconductors show zero electrical resistance and the Meissner effect, supporting specialised magnets rather than ordinary room-temperature wiring claims. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q56. Which option avoids the standard UPSC close-option trap about Circuits electrical power and safety?

A. Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena. B. Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability. C. CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public outcome are separate evidence rungs. D. Series branches share current and parallel branches share voltage; domestic loads are connected in parallel. Electrical power can be written VI under the relevant circuit conditions, transmission loss rises with I squared R, and fuse or MCB protection and earthing perform different safety functions. Below a critical temperature, superconductors show zero electrical resistance and the Meissner effect, supporting specialised magnets rather than ordinary room-temperature wiring claims.

Answer: D. Explanation: Series branches share current and parallel branches share voltage; domestic loads are connected in parallel. Electrical power can be written VI under the relevant circuit conditions, transmission loss rises with I squared R, and fuse or MCB protection and earthing perform different safety functions. Below a critical temperature, superconductors show zero electrical resistance and the Meissner effect, supporting specialised magnets rather than ordinary room-temperature wiring claims. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q57. Which statement correctly identifies Magnetism induction motors generators and transformers?

A. Changing magnetic flux induces an emf; generators convert mechanical to electrical energy, motors convert electrical to mechanical energy, and transformers use mutual induction to change AC voltage. A steady DC supply does not provide the changing flux needed for ordinary transformer action. B. Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena. C. Alpha radiation is a helium nucleus, beta commonly involves an electron or positron, and gamma is electromagnetic radiation; alpha is strongly ionising but weakly penetrating relative to gamma, whose penetration is much greater. Half-life is a nuclear decay property unaffected by ordinary temperature or pressure and supports radioisotope dating such as carbon-14, while fission splits heavy nuclei and fusion joins light nuclei. D. Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to brightness, treated as interchangeable or extended to every particle.

Answer: A. Explanation: Changing magnetic flux induces an emf; generators convert mechanical to electrical energy, motors convert electrical to mechanical energy, and transformers use mutual induction to change AC voltage. A steady DC supply does not provide the changing flux needed for ordinary transformer action. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q58. Which option preserves the technical boundary of Magnetism induction motors generators and transformers?

A. Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability. B. Changing magnetic flux induces an emf; generators convert mechanical to electrical energy, motors convert electrical to mechanical energy, and transformers use mutual induction to change AC voltage. A steady DC supply does not provide the changing flux needed for ordinary transformer action. C. Alpha radiation is a helium nucleus, beta commonly involves an electron or positron, and gamma is electromagnetic radiation; alpha is strongly ionising but weakly penetrating relative to gamma, whose penetration is much greater. Half-life is a nuclear decay property unaffected by ordinary temperature or pressure and supports radioisotope dating such as carbon-14, while fission splits heavy nuclei and fusion joins light nuclei. D. Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena.

Answer: B. Explanation: Changing magnetic flux induces an emf; generators convert mechanical to electrical energy, motors convert electrical to mechanical energy, and transformers use mutual induction to change AC voltage. A steady DC supply does not provide the changing flux needed for ordinary transformer action. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q59. Which statement uses Magnetism induction motors generators and transformers without changing its institution, unit or status?

A. CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public outcome are separate evidence rungs. B. Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena. C. Changing magnetic flux induces an emf; generators convert mechanical to electrical energy, motors convert electrical to mechanical energy, and transformers use mutual induction to change AC voltage. A steady DC supply does not provide the changing flux needed for ordinary transformer action. D. Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability.

Answer: C. Explanation: Changing magnetic flux induces an emf; generators convert mechanical to electrical energy, motors convert electrical to mechanical energy, and transformers use mutual induction to change AC voltage. A steady DC supply does not provide the changing flux needed for ordinary transformer action. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q60. Which option avoids the standard UPSC close-option trap about Magnetism induction motors generators and transformers?

A. The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time. B. CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public outcome are separate evidence rungs. C. Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability. D. Changing magnetic flux induces an emf; generators convert mechanical to electrical energy, motors convert electrical to mechanical energy, and transformers use mutual induction to change AC voltage. A steady DC supply does not provide the changing flux needed for ordinary transformer action.

Answer: D. Explanation: Changing magnetic flux induces an emf; generators convert mechanical to electrical energy, motors convert electrical to mechanical energy, and transformers use mutual induction to change AC voltage. A steady DC supply does not provide the changing flux needed for ordinary transformer action. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q61. Which statement correctly identifies Atomic quantum photoelectric and Bose boundary?

A. Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to brightness, treated as interchangeable or extended to every particle. B. Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena. C. Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability. D. Alpha radiation is a helium nucleus, beta commonly involves an electron or positron, and gamma is electromagnetic radiation; alpha is strongly ionising but weakly penetrating relative to gamma, whose penetration is much greater. Half-life is a nuclear decay property unaffected by ordinary temperature or pressure and supports radioisotope dating such as carbon-14, while fission splits heavy nuclei and fusion joins light nuclei.

Answer: A. Explanation: Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to

brightness, treated as interchangeable or extended to every particle. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

Q62. Which option preserves the technical boundary of Atomic quantum photoelectric and Bose boundary?

A. CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public outcome are separate evidence rungs. B. Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to brightness, treated as interchangeable or extended to every particle. C. Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena. D. Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability.

Answer: B. Explanation: Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to brightness, treated as interchangeable or extended to every particle. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q63. Which statement uses Atomic quantum photoelectric and Bose boundary without changing its institution, unit or status?

A. CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public outcome are separate evidence rungs. B. Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability. C. Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to brightness, treated as interchangeable or extended to every particle. D. The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time.

**Answer: C. Explanation: Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while

Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to brightness, treated as interchangeable or extended to every particle. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q64. Which option avoids the standard UPSC close-option trap about Atomic quantum photoelectric and Bose boundary?

A. CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public outcome are separate evidence rungs. B. Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation. C. The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time. D. Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to brightness, treated as interchangeable or extended to every particle.

Answer: D. Explanation: Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to brightness, treated as interchangeable or extended to every particle. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q65. Which statement correctly identifies Nuclear radioactivity fission and fusion?

A. Alpha radiation is a helium nucleus, beta commonly involves an electron or positron, and gamma is electromagnetic radiation; alpha is strongly ionising but weakly penetrating relative to gamma, whose penetration is much greater. Half-life is a nuclear decay property unaffected by ordinary temperature or pressure and supports radioisotope dating such as carbon-14, while fission splits heavy nuclei and fusion joins light nuclei. B. Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability. C. Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena. D. CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public outcome are separate evidence rungs.

Answer: A. Explanation: Alpha radiation is a helium nucleus, beta commonly involves an electron or positron, and gamma is electromagnetic radiation; alpha is strongly ionising but weakly penetrating relative to gamma, whose penetration is much greater. Half-life is a nuclear decay property unaffected by ordinary temperature or pressure and supports radioisotope dating such as carbon-14, while fission splits heavy nuclei and fusion

joins light nuclei. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q66. Which option preserves the technical boundary of Nuclear radioactivity fission and fusion?

A. The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time. B. Alpha radiation is a helium nucleus, beta commonly involves an electron or positron, and gamma is electromagnetic radiation; alpha is strongly ionising but weakly penetrating relative to gamma, whose penetration is much greater. Half-life is a nuclear decay property unaffected by ordinary temperature or pressure and supports radioisotope dating such as carbon-14, while fission splits heavy nuclei and fusion joins light nuclei. C. CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public outcome are separate evidence rungs. D. Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability.

Answer: B. Explanation: Alpha radiation is a helium nucleus, beta commonly involves an electron or positron, and gamma is electromagnetic radiation; alpha is strongly ionising but weakly penetrating relative to gamma, whose penetration is much greater. Half-life is a nuclear decay property unaffected by ordinary temperature or pressure and supports radioisotope dating such as carbon-14, while fission splits heavy nuclei and fusion joins light nuclei. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q67. Which statement uses Nuclear radioactivity fission and fusion without changing its institution, unit or status?

A. Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation. B. CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public outcome are separate evidence rungs. C. Alpha radiation is a helium nucleus, beta commonly involves an electron or positron, and gamma is electromagnetic radiation; alpha is strongly ionising but weakly penetrating relative to gamma, whose penetration is much greater. Half-life is a nuclear decay property unaffected by ordinary temperature or pressure and supports radioisotope dating such as carbon-14, while fission splits heavy nuclei and fusion joins light nuclei. D. The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time.

Answer: C. Explanation: Alpha radiation is a helium nucleus, beta commonly involves an electron or positron, and gamma is electromagnetic radiation; alpha is strongly ionising but weakly penetrating relative to gamma, whose penetration is much greater. Half-life is a nuclear decay property unaffected by ordinary temperature or pressure and supports radioisotope dating such as carbon-14, while fission splits heavy nuclei and fusion joins light nuclei. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q68. Which option avoids the standard UPSC close-option trap about Nuclear radioactivity fission and fusion?

A. Newton's first law defines inertia, the second links net force to the rate of change of momentum and becomes F = ma for constant mass, and the third pairs equal and opposite forces on different bodies; impulse equals change in momentum and explains why increasing stopping time can reduce average force. B. Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation. C. The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time. D. Alpha radiation is a helium nucleus, beta commonly involves an electron or positron, and gamma is electromagnetic radiation; alpha is strongly ionising but weakly penetrating relative to gamma, whose penetration is much greater. Half-life is a nuclear decay property unaffected by ordinary temperature or pressure and supports radioisotope dating such as carbon-14, while fission splits heavy nuclei and fusion joins light nuclei.

Answer: D. Explanation: Alpha radiation is a helium nucleus, beta commonly involves an electron or positron, and gamma is electromagnetic radiation; alpha is strongly ionising but weakly penetrating relative to gamma, whose penetration is much greater. Half-life is a nuclear decay property unaffected by ordinary temperature or pressure and supports radioisotope dating such as carbon-14, while fission splits heavy nuclei and fusion joins light nuclei. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q69. Which statement correctly identifies Relativity gravitational waves and astronomy?

A. Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena. B. The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time. C. CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public outcome are separate evidence rungs. D. Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability.

Answer: A. Explanation: Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q70. Which option preserves the technical boundary of Relativity gravitational waves and astronomy?

A. The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time. B. Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena. C. CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public outcome are separate evidence rungs. D. Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation.

Answer: B. Explanation: Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q71. Which statement uses Relativity gravitational waves and astronomy without changing its institution, unit or status?

A. The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time. B. Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation. C. Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena. D. Newton's first law defines inertia, the second links net force to the rate of change of momentum and becomes $F = ma$ for constant mass, and the third pairs equal and opposite forces on different bodies; impulse equals change in momentum and explains why increasing stopping time can reduce average force.

Answer: C. Explanation: Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q72. Which option avoids the standard UPSC close-option trap about Relativity gravitational waves and astronomy?

A. Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation. B. Work transfers energy when force has a displacement component along its direction; the work-energy theorem links net work to change in kinetic energy, while power is the rate of doing work or transferring energy. Force, energy and power therefore have different dimensions and cannot be interchanged. C. Newton's first law defines inertia, the second links net force to the rate of change of momentum and becomes $F = ma$ for constant mass, and the third pairs equal and opposite forces on different bodies; impulse equals change in momentum and explains why increasing stopping time can reduce average force. D. Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena.

Answer: D. Explanation: Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q73. Which statement correctly identifies Semiconductors LEDs lasers radar and VLC?

A. Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability. B. Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation. C. The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time. D. CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public outcome are separate evidence rungs.

Answer: A. Explanation: Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q74. Which option preserves the technical boundary of Semiconductors LEDs lasers radar and VLC?

A. The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time. B. Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability. C. Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation. D. Newton's first law defines inertia, the second links net force to the rate of change of momentum and becomes F = ma for constant mass, and the third pairs equal and opposite forces on different bodies; impulse equals change in momentum and explains why increasing stopping time can reduce average force.

Answer: B. Explanation: Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q75. Which statement uses Semiconductors LEDs lasers radar and VLC without changing its institution, unit or status?

A. Newton's first law defines inertia, the second links net force to the rate of change of momentum and becomes F = ma for constant mass, and the third pairs equal and opposite forces on different bodies; impulse equals change in momentum and explains why increasing stopping time can reduce average force. B. Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation. C. Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability. D. Work transfers energy when force has a displacement component along its direction; the work-energy theorem links net work to change in kinetic energy, while power is the rate of doing work or transferring energy. Force, energy and power therefore have different dimensions and cannot be interchanged.

Answer: C. Explanation: Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q76. Which option avoids the standard UPSC close-option trap about Semiconductors LEDs lasers radar and VLC?

A. Work transfers energy when force has a displacement component along its direction; the work-energy theorem links net work to change in kinetic energy, while power is the rate of doing work or transferring energy. Force, energy and power therefore have different dimensions and cannot be interchanged. B. Gravity supplies the centripetal acceleration of an orbiting body; orbital weightlessness is continuous free fall, not absence of gravity. The Basic owner gives Earth's escape velocity as approximately 11.2 km/s and a geostationary orbit as approximately 35,786 km above the equator with one sidereal-day period; escape velocity is independent of projectile mass, while polar or sun-synchronous orbits occupy a different plane and application class. C. Newton's first law defines inertia, the second links net force to the rate of change of momentum and becomes F = ma for constant mass, and the third pairs equal and opposite forces on different bodies; impulse equals change in momentum and explains why increasing stopping time can reduce average force. D. Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability.

Answer: D. Explanation: Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q77. Which statement correctly identifies Metrology recorder and status firewall?

A. CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public outcome are separate evidence rungs. B. Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation. C. The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time. D. Newton's first law defines inertia, the second links net force to the rate of change of momentum and becomes F = ma for constant mass, and the third pairs equal and opposite forces on different bodies; impulse equals change in momentum and explains why increasing stopping time can reduce average force.

Answer: A. Explanation: CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public outcome are separate evidence rungs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q78. Which option preserves the technical boundary of Metrology recorder and status firewall?

A. Newton's first law defines inertia, the second links net force to the rate of change of momentum and becomes F = ma for constant mass, and the third pairs equal and opposite forces on different bodies; impulse equals change in momentum and explains why increasing stopping time can reduce average force. B. CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public outcome are separate evidence rungs. C. Work transfers energy when force has a displacement component along its direction; the work-energy theorem links net work to change in kinetic energy, while power is the rate of doing work or transferring energy. Force, energy and power therefore have different dimensions and cannot be interchanged. D. Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation.

Answer: B. Explanation: CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public outcome are separate evidence rungs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q79. Which statement uses Metrology recorder and status firewall without changing its institution, unit or status?

A. Newton's first law defines inertia, the second links net force to the rate of change of momentum and becomes F = ma for constant mass, and the third pairs equal and opposite forces on different bodies; impulse equals change in momentum and explains why increasing stopping time can reduce average force. B. Gravity supplies the centripetal acceleration of an orbiting body; orbital weightlessness is continuous free fall, not absence of gravity. The Basic owner gives Earth's escape velocity as approximately 11.2 km/s and a geostationary orbit as approximately 35,786 km above the equator with one sidereal-day period; escape velocity is independent of projectile mass, while polar or sun-synchronous orbits occupy a different plane and application class. C. CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public outcome are separate evidence rungs. D. Work transfers energy when force has a displacement component along its direction; the work-energy theorem links net work to change in kinetic energy, while power is the rate of doing work or transferring energy. Force, energy and power therefore have different dimensions and cannot be interchanged.

Answer: C. Explanation: CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public outcome are separate evidence rungs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q80. Which option avoids the standard UPSC close-option trap about Metrology recorder and status firewall?

A. Work transfers energy when force has a displacement component along its direction; the work-energy theorem links net work to change in kinetic energy, while power is the rate of doing work or transferring energy. Force, energy and power therefore have different dimensions and cannot be interchanged. B. Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude. C. Gravity supplies the centripetal acceleration of an orbiting body; orbital weightlessness is continuous free fall, not absence of gravity. The Basic owner gives Earth's escape velocity as approximately 11.2 km/s and a geostationary orbit as approximately 35,786 km above the equator with one sidereal-day period; escape velocity is independent of projectile mass, while polar or sun-synchronous orbits occupy a different plane and application class. D. CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public outcome are separate evidence rungs.

Answer: D. Explanation: CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public outcome are separate evidence rungs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

Audited ledgers route eleven Prelims demands from 2018-2026 and two GS-III demands from 2018 and 2021 to this owner: relativity predictions; gravitational waves and black-hole merger; VLC; pressure-cooker physics; sodium lamps versus LEDs; light-year measurement; short-range wireless classification; accelerometers; Cepheids, nebulae and pulsars; radar uses; aircraft black-box recorders and underwater detection; Bose-Einstein statistics; and blue-LED impact. Three representative cards preserve the historical, recent and Mains routes without reproducing or inferring any objective answer key.

PYQ DEMAND CARD 1 - 2018-2023 Prelims GS-I

Demand: Assess the routed distinctions involving pressure-cooker physics, light-year measurement, accelerometer functions, astronomical objects, VLC, wireless classification and gravitational-wave observations.

Status: Representative historical objective card spanning routed physics concepts; official 2018-2023 keys are unavailable locally, so no option, answer letter or objective key is asserted.

Model solution: SI units and dimensional boundary: The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time. **Motion acceleration and measurement:** Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation. **Pressure buoyancy and fluid principles:** Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude. **Electromagnetic spectrum and communication boundary:** In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence. **Relativity gravitational waves and astronomy:** Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

PYQ DEMAND CARD 2 - 2024 and 2026 Prelims GS-I

Demand: Assess the routed uses and boundaries of radar, aircraft black-box recorders, underwater detection and crash-survivable memory.

Status: Representative recent objective card covering 2024 Q34 and 2026 Q44; the 2024 official key and provisional 2026 key are not reproduced, and no answer is inferred.

Model solution: Electromagnetic spectrum and communication boundary: In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence. **Metrology recorder and status firewall:** CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public outcome are separate evidence rungs. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

PYQ DEMAND CARD 3 - 2018 and 2021 GS-III

Demand: Discuss the contribution of Bose-Einstein statistics to physics and explain the everyday-life impact of the blue LED invention recognised by the 2014 Nobel Prize.

Status: Representative routed Mains card covering 2018 Q5 at 10 marks and 2021 Q16 at 15 marks; it supplies an evidence-bounded answer route, not an official model answer.

Model solution: Atomic quantum photoelectric and Bose boundary: Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to brightness, treated as interchangeable or extended to every particle. **Semiconductors LEDs lasers radar and VLC:** Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish SI units, dimensions, scalar-vector quantities and measurement limits. Answer in about 150 words.

Model thesis: Claim: SI units and dimensional boundary. **Named evidence/example:** The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Motion acceleration and measurement. **Named evidence/example:** Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time.
- Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation.

Qualified conclusion: Claim: SI units and dimensional boundary. **Named evidence/example:** The SI uses metre, kilogram, second, ampere, kelvin, mole and candela as base units; derived units express products of powers of base units, including hertz as inverse second, joule as kilogram metre squared per second squared and watt as kilogram metre squared per second cubed. Dimensional analysis can test consistency but cannot prove that a physical equation, measurement or causal claim is correct; a light-year is a unit of distance, not time. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Motion acceleration and measurement. **Named evidence/example:** Distance and speed are scalars, while displacement, velocity and acceleration require direction; an accelerometer senses acceleration and can support motion, tilt, free-fall or crash detection, but a sensor reading still requires calibration, orientation and system interpretation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

ORIGINAL MAINS 2 - 10 MARKS

Question: Explain how Newton's laws, momentum, impulse, work, energy and power illuminate everyday technology. Answer in about 150 words.

Model thesis: Claim: Force momentum impulse and Newton laws. **Named evidence/example:** Newton's first law defines inertia, the second links net force to the rate of change of momentum and becomes F = ma for constant mass, and the third pairs equal and opposite forces on different bodies; impulse equals change in momentum and explains why increasing stopping time can reduce average force. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Work energy and power. **Named evidence/example:** Work transfers energy when force has a displacement component along its direction; the work-energy theorem links net work to change in kinetic energy, while power is the rate of doing work or transferring energy. Force, energy and power therefore have different dimensions and cannot be interchanged. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Newton's first law defines inertia, the second links net force to the rate of change of momentum and becomes F = ma for constant mass, and the third pairs equal and opposite forces on different bodies; impulse equals change in momentum and explains why increasing stopping time can reduce average force.
- Work transfers energy when force has a displacement component along its direction; the work-energy theorem links net work to change in kinetic energy, while power is the rate of doing work or transferring energy. Force, energy and power therefore have different dimensions and cannot be interchanged.

Qualified conclusion: Claim: Force momentum impulse and Newton laws. **Named evidence/example:** Newton's first law defines inertia, the second links net force to the rate of change of momentum and becomes F = ma for constant mass, and the third pairs equal and opposite forces on different bodies; impulse equals change in momentum and explains why increasing stopping time can reduce average force. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Work energy and power. **Named evidence/example:** Work transfers energy when force has a displacement component along its direction; the work-energy theorem links net work to change in kinetic energy, while power is the rate of doing work or transferring energy. Force, energy and power therefore have different dimensions and cannot be interchanged. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

ORIGINAL MAINS 3 - 15 MARKS

Question: Analyse gravitation, orbital motion, pressure, buoyancy and fluid principles through applications and traps. Answer in about 250 words.

Model thesis: Claim: Gravitation circular motion and orbits. **Named evidence/example:** Gravity supplies the centripetal acceleration of an orbiting body; orbital weightlessness is continuous free fall, not absence of gravity. The Basic owner gives Earth's escape velocity as approximately 11.2 km/s and a geostationary orbit as approximately 35,786 km above the equator with one sidereal-day period; escape velocity is independent of projectile mass, while polar or sun-synchronous orbits occupy a different plane and application class. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Pressure buoyancy and fluid principles. **Named evidence/example:** Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Gravity supplies the centripetal acceleration of an orbiting body; orbital weightlessness is continuous free fall, not absence of gravity. The Basic owner gives Earth's escape velocity as approximately 11.2 km/s and a geostationary orbit as approximately 35,786 km above the equator with one sidereal-day period; escape velocity is independent of projectile mass, while polar or sun-synchronous orbits occupy a different plane and application class.
- Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its

assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude.

Qualified conclusion: Claim: Gravitation circular motion and orbits. **Named evidence/example:** Gravity supplies the centripetal acceleration of an orbiting body; orbital weightlessness is continuous free fall, not absence of gravity. The Basic owner gives Earth's escape velocity as approximately 11.2 km/s and a geostationary orbit as approximately 35,786 km above the equator with one sidereal-day period; escape velocity is independent of projectile mass, while polar or sun-synchronous orbits occupy a different plane and application class. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Pressure buoyancy and fluid principles. **Named evidence/example:** Pressure is force per unit area; Pascal's law transmits applied pressure in a confined fluid, Archimedes' principle links buoyant force to the weight of displaced fluid, and Bernoulli's principle relates pressure and flow under its assumptions. Surface tension shapes droplets and capillarity supports liquid rise in narrow spaces; a pressure cooker raises boiling temperature by raising pressure, whereas lower atmospheric pressure lowers boiling temperature at altitude. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

ORIGINAL MAINS 4 - 15 MARKS

Question: Discuss heat, thermodynamics, waves, sound and optics as linked but distinct physical systems. Answer in about 250 words.

Model thesis: Claim: Heat temperature specific and latent heat. **Named evidence/example:** Temperature describes thermal state and is not heat; heat is energy transferred because of a temperature difference. Specific heat compares energy required for a temperature rise, while latent heat is absorbed or released during a phase change without a temperature change. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Thermodynamics and heat transfer. **Named evidence/example:** Conduction, convection and radiation are distinct heat-transfer modes, and radiation needs no material medium. The first law applies energy conservation to thermal systems; the second law fixes spontaneous heat flow from hotter to colder bodies and rules out a perfectly efficient heat engine. An ideal black body absorbs and emits across wavelengths, and hotter bodies shift peak emission toward shorter wavelengths. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Wave relation and sound. **Named evidence/example:** A wave transfers energy without net transport of matter, with speed equal to frequency multiplied by wavelength. Sound is mechanical and travels fastest in solids and slowest in gases; frequency primarily sets pitch, amplitude affects loudness, Doppler shift follows relative motion, and resonance amplifies forced oscillation near natural frequency. Infrasound is below 20 Hz, audible sound spans about 20 Hz to 20 kHz and ultrasound is above 20 kHz; ultrasound must not be confused with supersonic motion, while echo and reverberation differ by time separation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Reflection mirrors and image logic. **Named evidence/example:** Reflection obeys equality of incidence and reflection angles; plane, concave and convex mirrors form images according to geometry and object position. Mirror action is reflection, so it must not be explained through the refraction mechanism used by lenses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Refraction total internal reflection and lenses. **Named evidence/example:** Refraction follows a change in light speed across a boundary while frequency remains unchanged; total internal reflection requires travel from denser to rarer medium beyond the critical angle and underpins optical fibres. Convex and concave lenses converge and diverge respectively, with myopia corrected by a concave lens, hypermetropia by a convex lens, presbyopia commonly by bifocals and astigmatism by cylindrical lenses. Dispersion separates colours, Rayleigh scattering favours shorter wavelengths, Mie scattering acts with larger particles and the Tyndall effect reveals colloids. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Temperature describes thermal state and is not heat; heat is energy transferred because of a temperature difference. Specific heat compares energy required for a temperature rise, while latent heat is absorbed or released during a phase change without a temperature change.
- Conduction, convection and radiation are distinct heat-transfer modes, and radiation needs no material medium. The first law applies energy conservation to thermal systems; the second law fixes spontaneous heat flow from hotter to colder bodies and rules out a perfectly efficient heat engine. An ideal black body absorbs and emits across wavelengths, and hotter bodies shift peak emission toward shorter wavelengths.
- A wave transfers energy without net transport of matter, with speed equal to frequency multiplied by wavelength. Sound is mechanical and travels fastest in solids and slowest in gases; frequency primarily sets pitch, amplitude affects loudness, Doppler shift follows relative motion, and resonance amplifies forced oscillation near natural frequency. Infrasound is below 20 Hz, audible sound spans about 20 Hz to 20 kHz and ultrasound is above 20 kHz; ultrasound must not be confused with supersonic motion, while echo and reverberation differ by time separation.
- Reflection obeys equality of incidence and reflection angles; plane, concave and convex mirrors form images according to geometry and object position. Mirror action is reflection, so it must not be explained through the refraction mechanism used by lenses.
- Refraction follows a change in light speed across a boundary while frequency remains unchanged; total internal reflection requires travel from denser to rarer medium beyond the critical angle and underpins optical fibres. Convex and concave lenses converge and diverge respectively, with myopia corrected by a concave lens, hypermetropia by a convex lens, presbyopia commonly by bifocals and astigmatism by cylindrical lenses. Dispersion separates colours, Rayleigh scattering favours shorter wavelengths, Mie scattering acts with larger particles and the Tyndall effect reveals colloids.

Qualified conclusion: Claim: Heat temperature specific and latent heat. **Named evidence/example:** Temperature describes thermal state and is not heat; heat is energy transferred because of a temperature difference. Specific heat compares energy required for a temperature rise, while latent heat is absorbed or released during a phase change without a temperature change. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Thermodynamics and heat transfer. **Named evidence/example:** Conduction, convection and radiation are distinct heat-transfer modes, and radiation needs no material medium. The first law applies energy conservation to thermal systems; the second law fixes spontaneous heat flow from hotter to colder bodies and rules out a perfectly efficient heat engine. An ideal black body absorbs and emits across wavelengths, and hotter bodies shift peak emission toward shorter wavelengths. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Wave relation and sound. **Named evidence/example:** A wave transfers energy without net transport of matter, with speed equal to frequency multiplied by wavelength. Sound is mechanical and travels fastest in solids and slowest in gases; frequency primarily sets pitch, amplitude affects loudness, Doppler shift follows relative motion, and resonance amplifies forced oscillation near natural frequency. Infrasound is below 20 Hz, audible sound spans about 20 Hz to 20 kHz and ultrasound is above 20 kHz; ultrasound must not be confused with supersonic motion, while echo and reverberation differ by time separation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Reflection mirrors and image logic. **Named evidence/example:** Reflection obeys equality of incidence and reflection angles; plane, concave and convex mirrors form images according to geometry and object position. Mirror action is reflection, so it must not be explained through the refraction mechanism used by lenses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Refraction total internal reflection and lenses. **Named evidence/example:** Refraction follows a change in light speed across a boundary while frequency remains unchanged; total internal reflection requires travel from denser to rarer medium beyond the critical angle and underpins optical fibres. Convex and concave lenses converge and diverge respectively, with myopia corrected by a concave lens, hypermetropia by a convex lens, presbyopia commonly by bifocals and astigmatism by cylindrical lenses. Dispersion separates colours, Rayleigh scattering favours shorter wavelengths, Mie scattering acts with larger particles and the Tyndall effect reveals colloids. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

ORIGINAL MAINS 5 - 20 MARKS

Question: Examine how electricity, magnetism and the electromagnetic spectrum underpin power, communication and safety systems. Answer in about 300 words.

Model thesis: Claim: Electromagnetic spectrum and communication boundary. **Named evidence/example:** In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Current voltage resistance and Ohm boundary. **Named evidence/example:** Electric current is charge flow per unit time, potential difference is work per unit charge and resistance opposes current. Ohm's law $V = IR$ applies under stated physical conditions to ohmic conductors, so it is not a universal description of every material or device. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Circuits electrical power and safety. **Named evidence/example:** Series branches share current and parallel branches share voltage; domestic loads are connected in parallel. Electrical power can be written VI under the relevant circuit conditions, transmission loss rises with I squared R, and fuse or MCB protection and earthing perform different safety functions. Below a critical temperature, superconductors show zero electrical resistance and the Meissner effect, supporting specialised magnets rather than ordinary room-temperature wiring claims. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Magnetism induction motors generators and transformers. **Named evidence/example:** Changing magnetic flux induces an emf; generators convert mechanical to electrical energy, motors convert electrical to mechanical energy, and transformers use mutual induction to change AC voltage. A steady DC supply does not provide the changing flux needed for ordinary transformer action. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence.
- Electric current is charge flow per unit time, potential difference is work per unit charge and resistance opposes current. Ohm's law $V = IR$ applies under stated physical conditions to ohmic conductors, so it is not a universal

description of every material or device.

- Series branches share current and parallel branches share voltage; domestic loads are connected in parallel. Electrical power can be written VI under the relevant circuit conditions, transmission loss rises with I squared R, and fuse or MCB protection and earthing perform different safety functions. Below a critical temperature, superconductors show zero electrical resistance and the Meissner effect, supporting specialised magnets rather than ordinary room-temperature wiring claims.
- Changing magnetic flux induces an emf; generators convert mechanical to electrical energy, motors convert electrical to mechanical energy, and transformers use mutual induction to change AC voltage. A steady DC supply does not provide the changing flux needed for ordinary transformer action.

Qualified conclusion: Claim: Electromagnetic spectrum and communication boundary. **Named evidence/example:** In increasing frequency the electromagnetic spectrum runs radio, microwave, infrared, visible, ultraviolet, X-rays and gamma rays; all travel at light speed in vacuum but interact differently with matter. Microwaves, infrared thermal sensing and ionising X-ray or gamma uses require band-specific treatment; radar uses radio waves, visible-light communication uses modulated visible light, and short-range wireless labels require technology-specific range and protocol evidence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Current voltage resistance and Ohm boundary. **Named evidence/example:** Electric current is charge flow per unit time, potential difference is work per unit charge and resistance opposes current. Ohm's law V = IR applies under stated physical conditions to ohmic conductors, so it is not a universal description of every material or device. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Circuits electrical power and safety. **Named evidence/example:** Series branches share current and parallel branches share voltage; domestic loads are connected in parallel. Electrical power can be written VI under the relevant circuit conditions, transmission loss rises with I squared R, and fuse or MCB protection and earthing perform different safety functions. Below a critical temperature, superconductors show zero electrical resistance and the Meissner effect, supporting specialised magnets rather than ordinary room-temperature wiring claims. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Magnetism induction motors generators and transformers. **Named evidence/example:** Changing magnetic flux induces an emf; generators convert mechanical to electrical energy, motors convert electrical to mechanical energy, and transformers use mutual induction to change AC voltage. A steady DC supply does not provide the changing flux needed for ordinary transformer action. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

ORIGINAL MAINS 6 - 20 MARKS

Question: Evaluate how atomic, nuclear and relativistic physics becomes technology through semiconductors, radiation, instruments and evidence-status discipline. Answer in about 300 words.

Model thesis: Claim: Atomic quantum photoelectric and Bose boundary. **Named evidence/example:** Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to brightness, treated as interchangeable or extended to every particle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Nuclear radioactivity fission and fusion. **Named evidence/example:** Alpha radiation is a helium nucleus, beta commonly involves an electron or positron, and gamma is electromagnetic radiation; alpha is strongly ionising but weakly penetrating relative to gamma, whose penetration is much greater. Half-life is a nuclear decay property unaffected by ordinary temperature or pressure and supports radioisotope dating such as carbon-14, while fission splits heavy nuclei and fusion joins light nuclei. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Relativity gravitational waves and astronomy. **Named evidence/example:** Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Semiconductors LEDs lasers radar and VLC. **Named evidence/example:** Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Metrology recorder and status firewall. **Named evidence/example:** CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public outcome are separate evidence rungs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to brightness, treated as interchangeable or extended to every particle.
- Alpha radiation is a helium nucleus, beta commonly involves an electron or positron, and gamma is electromagnetic radiation; alpha is strongly ionising but weakly penetrating relative to gamma, whose penetration is much greater. Half-life is a nuclear decay property unaffected by ordinary temperature or pressure and supports radioisotope dating such as carbon-14, while fission splits heavy nuclei and fusion joins light nuclei.
- Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena.
- Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability.
- CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public

outcome are separate evidence rungs.

Qualified conclusion: Claim: Atomic quantum photoelectric and Bose boundary. **Named evidence/example:** Atomic and quantum physics require discrete states and particle-wave behaviour beyond classical mechanics; the photoelectric effect requires light above a threshold frequency, while Bose-Einstein statistics applies to indistinguishable integer-spin particles that may share a quantum state. Antimatter, neutrinos and the Higgs field are distinct modern-physics concepts; none should be reduced to brightness, treated as interchangeable or extended to every particle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Nuclear radioactivity fission and fusion. **Named evidence/example:** Alpha radiation is a helium nucleus, beta commonly involves an electron or positron, and gamma is electromagnetic radiation; alpha is strongly ionising but weakly penetrating relative to gamma, whose penetration is much greater. Half-life is a nuclear decay property unaffected by ordinary temperature or pressure and supports radioisotope dating such as carbon-14, while fission splits heavy nuclei and fusion joins light nuclei. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Relativity gravitational waves and astronomy. **Named evidence/example:** Special relativity sets light speed as the information-speed limit and relates mass and energy, while general relativity describes gravitation through spacetime geometry and predicts effects including gravitational waves. A gravitational-wave observation or black-hole merger inference depends on instruments and analysis; Cepheids, nebulae and pulsars are distinct astronomical objects or phenomena. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Semiconductors LEDs lasers radar and VLC. **Named evidence/example:** Pentavalent doping produces n-type material with electrons as majority carriers, trivalent doping produces p-type material with holes as majority carriers, and a p-n junction enables rectification, LEDs and photovoltaic behaviour. A bipolar transistor uses base current while a MOSFET uses an insulated gate field to control conduction; lasers rely on stimulated emission, coherence, near-monochromaticity and directionality, radar uses radio waves, and VLC uses visible light. No device label alone establishes efficiency, range or suitability. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Metrology recorder and status firewall. **Named evidence/example:** CSIR-NPL anchors measurement standards and metrology; CSIR-CEERI supports electronics and device research, while ISM, the National Quantum Mission and the DAE or BARC ecosystem connect physics to policy and application. Accelerometers, radar and crash-survivable flight recorders perform different sensing, detection or preservation functions, and underwater recovery adds a separate location problem. A standard, laboratory result, announced scheme, scheduled event, detected signal, calibrated reading, operational device and verified public outcome are separate evidence rungs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.